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SELECTION OF EXPERTS ENERGY BASED MODEL GADGET

Abstract

A thermodynamic selection of experts energy-based model gadget includes a SoftMax gadget, a set of oscillators having a potential which is used to modify input data, and multiple energy-based models for processing data. The SoftMax gadget may produce one-hot encoded vectors which may be used by the set of oscillators having a potential which is used to modify input data, such that the modified input data corresponds to one of the multiple energy-based models for processing data.

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Background/Summary

RELATED APPLICATION [0001] This application claims benefit of priority to U.S. Provisional Application Ser. No. 63/562,573, entitled “SELECTION OF EXPERTS ENERGY BASED MODEL GADGET,” filed Mar. 7, 2024, and which is incorporated herein by reference in its entirety.

BACKGROUND
Description of Related Art
[0002] Various algorithms, such as machine learning algorithms, often use statistical probabilities to make decisions or to model systems. Some such learning algorithms may use Bayesian statistics, or may use other statistical models that have a theoretical basis in natural phenomena. Also, machine learning algorithms themselves may be implemented using Bayesian statistics, or may use other statistical models that have a theoretical basis in natural phenomena.
[0003] Generating such statistical probabilities may involve performing complex calculations which may require both time and energy to perform, thus increasing a latency of execution of the algorithm and/or negatively impacting energy efficiency. In some scenarios, calculation of such statistical probabilities using classical computing devices may result in non-trivial increases in execution time of algorithms and/or energy usage to execute such algorithms.
[0004] As an alternative, algorithms may be performed using thermodynamic computers. However, communication between multiple algorithms implemented on a thermodynamic computing device and/or communications between thermodynamic computing devices may require converting information into a classical computing device form, thus reducing at least some of the benefits of a thermodynamic computer implementation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1**A is a high-level block diagram of a thermodynamic chip arrangement for a selection of experts gadget, according to some embodiments.
[0006] FIG. **1**B is a block diagram of a set of SoftMax output oscillators in a particular one-hot vector configuration which selects a particular energy-based model, according to some embodiments.

[0007] FIG. **1**C is the high-level block diagram of the thermodynamic chip arrangement with the SoftMax output oscillators in the particular one-hot vector configuration, according to some embodiments.
[0008] FIG. **1**D is a block diagram of a configuration for an overall selection of experts gadget which produces results using a set of energy-based models, according to some embodiments.
[0009] FIG. **1**E is a block diagram of a configuration for a multiplication gadget which accepts an input vector and results from a gating model such as a SoftMax gadget in order to select an energy-based model from a set of energy-based models to process the input vector, according to some embodiments.
[0010] FIG. **2**A is a block diagram of oscillators which are storing input data and a SoftMax data sample and oscillators of a multiplication gadget at a first time (T.sub.1) such as prior to couplings between the oscillators being established, according to some embodiments.
[0011] FIG. **2**B is a block diagram of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a second time (T.sub.2), such as after couplings between the oscillators are established and while thermodynamic evolution is occurring, according to some embodiments.
[0012] FIG. **2**C is a block diagram of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a third time (T.sub.3), such as after thermodynamic equilibrium is reached and the multiplication gadget has a data sample representing an expectation value vector, according to some embodiments.
[0013] FIG. **2**D is a block diagram of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a third time (T.sub.3), such as such as after thermodynamic equilibrium is reached and a set of data sample receiver oscillators are able to receive the data sample representing the expectation value vector from the multiplication gadget, according to some embodiments.
[0014] FIG. **3** illustrates an example coupling for an analog SoftMax gadget, wherein ancilla oscillators (ϕ.sub.a.sub.j.sub.(l)) are used to emulate an all-to-all coupling between input/output oscillators (ϕ.sub.b.sub.j) of the analog SoftMax gadget, wherein the input/output oscillators (ϕ.sub.b.sub.j) and the additional oscillators (ϕ.sub.a.sub.j.sub.(l)) have a reduced degree of connectivity as compared to input/output oscillators (ϕ.sub.b.sub.j) used in an all-to-all coupling for a similar sized array of input/output oscillators, according to some embodiments.
[0015] FIG. **4** illustrates graphs of potentials for a given oscillator of the analog SoftMax gadget, wherein the given oscillator has a dual-well potential. FIG. **4** further illustrates how increasing the parameter A.sub.1 in an engineered potential for the analog SoftMax gadget causes the walls and intermediate barrier between the two wells of the dual-well potential to be more steep, such that the dual-well oscillator is more likely to evolve to a value of 0 or 1 as required by the engineered potential for the analog SoftMax gadget, according to some embodiments.
[0016] FIG. **5** illustrates an example attention block of a machine learning model, that may be implemented in an analog manner using one or more thermodynamic chips, wherein an analog selection of experts gadget is used at least in part to implement an addition and normalization layer, according to some embodiments.
[0017] FIG. **6** is a flowchart illustrating a process for implementing a selection of experts function using an analog selection of experts gadget, according to some embodiments.
[0018] FIG. **7**A illustrates additional details of a relay gadget implemented using a thermodynamic chip, wherein the relay gadget is configured to relay thermodynamic information between a first energy-based model (EBM), such as a SoftMax gadget, and a second energy-based model (EBM), such as an analog multiplication gadget, according to some embodiments.
[0019] FIG. **7**B is high-level diagram similar to FIG. **7**A, wherein the relay gadget does not include a bias oscillator, according to some embodiments.
[0020] FIG. **8** is a high-level flowchart illustrating a process of relaying thermodynamic information between an output oscillator, such as of a SoftMax gadget, and an input oscillator, such as an input oscillator of a multiplication gadget, according to some embodiments.
[0021] FIG. **9** is a high-level diagram illustrating an output oscillator, an input oscillator, and a relay gadget, wherein the relay gadget comprises a group of relay oscillators and is configured to relay expectation values of thermodynamic information between the output oscillator and the input oscillator, according to some embodiments.
[0022] FIG. **10** is a high-level diagram illustrating a spatial analogue relay gadget, wherein respective ones of relay oscillators of a group of relay oscillators are configured to store respective sample values of an output oscillator, according to some embodiments.
[0023] FIG. **11** is a high-level diagram illustrating a temporal analogue relay gadget, wherein a group of relay oscillators comprises a single relay oscillator, according to some embodiments.
[0024] FIG. **12** is a high-level diagram illustrating a series analogue relay gadget, wherein a group of relay oscillators comprises a plurality of relay oscillators arranged in series, according to some embodiments.
[0025] FIG. **13**A illustrates example couplings between visible neurons of an energy-based model (EBM), according to some embodiments.
[0026] FIG. **13**B illustrates example couplings between visible neurons and non-visible neurons (e.g., hidden neurons) of an energy-based model (EBM), according to some embodiments.
[0027] FIG. **14** is high-level diagram illustrating a process of determining weights and biases to be used in an energy-based model (EBM), wherein the weights and biases are determined using measurement values for synapse oscillators, according to some embodiments.
[0028] FIG. **15** is high-level diagram illustrating a process of determining weights and biases to be used in an energy-based model (EBM), wherein the weights and biases are computed using a classical computing device, according to some embodiments.
[0029] FIG. **16** is high-level diagram illustrating an example neuro-thermodynamic computer comprising a thermodynamic chip (e.g., that implements one or more energy-based models (EBMs), an analog SoftMax gadget, and a relay gadget) included in a dilution refrigerator and coupled to a classical computing device in an environment external to the dilution refrigerator, according to some embodiments.
[0030] FIG. **17** is high-level diagram illustrating an example neuro-thermodynamic computer comprising a thermodynamic chip (e.g., that implements one or more energy-based models (EBMs), an analog SoftMax gadget, and a relay gadget) included in a dilution refrigerator and coupled to a classical computing device that is also included in the dilution refrigerator, according to some embodiments.
[0031] FIG. **18** is high-level diagram illustrating an example neuro-thermodynamic computer comprising one or more thermodynamic chips (e.g., that implement one or more energy-based models (EBMs), an analog SoftMax gadget, and a relay gadget) coupled to a classical computing device in an environment other than a dilution refrigerator, according to some embodiments.
[0032] FIG. **19** is a high-level diagram illustrating oscillators included in a substrate of a thermodynamic chip and a mapping of the oscillators to logical neurons or synapses of the thermodynamic chip, according to some embodiments.
[0033] FIG. **20** is an additional high-level diagram illustrating oscillators included in a substrate of a thermodynamic chip mapped to logical neurons, weights, and biases (e.g., synapses) of a neuro-thermodynamic computing system, according to some embodiments.
[0034] FIG. **21** illustrates an example apparatus for measuring positions of oscillators of a thermodynamic chip using a flux read-out device, according to some embodiments.
[0035] FIG. **22** illustrates an example apparatus for measuring momentums of oscillators of a thermodynamic chip using a charge read-out device, according to some embodiments.
[0036] FIG. **23** is a block diagram illustrating an example computer system that may be used in at least some embodiments.
[0037] While embodiments are described herein by way of example for several embodiments and illustrative drawings, those skilled in the art will recognize that embodiments are not limited to the embodiments or drawings described. It should be understood, that the drawings and detailed

description thereto are not intended to limit embodiments to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope as defined by the appended claims. The headings used herein are for organizational purposes only and are not meant to be used to limit the scope of the description or the claims. As used throughout this application, the word “may” is used in a permissive sense (i.e., meaning having the potential to), rather than the mandatory sense (i.e., meaning must). Similarly, the words “include,” “including,” and “includes” mean including, but not limited to. When used in the claims, the term “or” is used as an inclusive or and not as an exclusive or. For example, the phrase “at least one of x, y, or z” means any one of x, y, and z, as well as any combination thereof.

DETAILED DESCRIPTION

[0038] The present disclosure relates to methods, systems, and an apparatus for processing data using a thermodynamic chip implementation of multiple energy-based models. The energy-based models that process data, which are called experts, are selected to process input data or modified input data based on the output of a SoftMax energy-based model gadget. For particular types of data, a particular expert of the set may be the best expert of the set of experts for processing the data. An analog SoftMax gadget and an analog multiplication gadget may be used in combination as parts of a selection of experts gadget to select expert energy-based models to process the data.

[0039] The selection of experts gadget may enable greater efficiency in a thermodynamic computer by reducing the training costs for deep learning by performing the function of a mixture model. The selection of experts gadget works with a gating function to select particular models based on input data. The gating function may be implemented by a SoftMax gadget or a similar selection gadget, and may take as input the initial input data or modified gating input data which has been modified from initial input data by an energy-based model as part of the gating function. The gating function may provide an indication of which energy-based model to which the multiplication gadget is to deliver the input data, which may be modified input data generated by combining the input data with the output of the gating gadget (i.e., the SoftMax gadget), which may use a gating function. In some embodiments, the gating function may provide weights that the selection of experts gadget may apply to the input data for the set of energy-based models which are to generate one or more outputs. The one or more outputs may be read out of the thermodynamic chip to a traditional computer or may be used in another energy-based model, for example, by being stored in relay gadgets.

[0040] Embodiments described herein further relate to performing computer operations using a thermodynamic chip and more specifically to relaying thermodynamic information between components, such as components of a neuro-thermodynamic computing device, while maintaining the information in a thermodynamic state. This can be contrasted with other approaches to communicate information that involve reading out thermodynamic information, such as using a classical computing device, and then relaying the information in classical form. For example, the ability to relay thermodynamic information directly between components in a neuro-thermodynamic computer avoids issues associated with readout to a classical computing device, such as read-out error, loss of information, and/or delays associated with performing readout. Moreover, if the information is to be used by another component of a neuro-thermodynamic computing device, relay of the information in a thermodynamic state avoids other delays such as would be incurred if required to initialize a receiving component to have an initial state corresponding to a state of the thermodynamic information that was read out from another component, wherein the relayed information is not already in a thermodynamic state. In some embodiments, such relay techniques as described herein may be used to relay thermodynamic information between energy-based models (EBMs). Such energy-based models (EBMs) may include trained models that evolve according to Langevin dynamics, and which may be used to generate inferences, such as machine learning (ML) inferences. For example, an ML model used to generate a ML inference may be physically implemented as a trained energy-based model (EBM).

[0041] An EBM may be an analog configuration of oscillators which may process and transfer information through connections to other oscillators. One type of information which the oscillators may use is the degrees of freedom of an oscillator, which may be represented by ¢. The EBMs may process the input data according to internal specifications. The internal specifications may be specific to the function of the EBMs, and may be specific to particular data types. A Hamiltonian or potential of an EBM may correspond to a function, such that an EBM with a particular Hamiltonian or potential is configured to execute the function associated with the Hamiltonian or potential. The EBMs may be analog devices implementing strategies used in deep learning for machine learning models, as are known for classical computing. A SoftMax gadget and a multiplication gadget are EBMs, and will be referred to by name to avoid confusion with the experts, which are the set of energy-based models that the SoftMax gadget and the multiplication gadget are selecting between.

[0042] Oscillators as used herein are described further at FIGS. **19**-**20**. A thermodynamic computer may use relay oscillators or relay gadgets to move information from one energy-based model to another. Relay oscillators and relay gadgets are described further at FIGS. **7**A-**12**.

[0043] In some embodiments, a neuro-thermodynamic processor may be configured such that learning algorithms for learning parameters of an energy-based model may be applied using Langevin dynamics. For example, as described herein, a thermodynamic chip of a neuro-thermodynamic processor may be configured such that, given a Hamiltonian that describes an energy-based model, weights and biases (e.g., synapses) may be calculated based on measurements taken from the thermodynamic chip as it naturally evolves according to Langevin dynamics. For example, a positive phase term, a negative phase term, and associated gradients needed to determine updated weights and biases for the energy-based model may be simply computed on an accompanying classical computing device, such as a field programmable gate array (FPGA) or application specific integrated circuit (ASIC), based on measurements taken from the oscillators of the thermodynamic chip. Such calculations performed on the accompanying classical computing device may be simple and non-complex as compared to other approaches that use the classical computing device to determine statistical probabilities (e.g., without using a thermodynamic chip). Also, in some embodiments, weights and biases used in an energy-based model may be determined iteratively, for example wherein a classical computing device is used to generate updated weights and biases, and wherein resulting inference performance is compared to training data to determine whether additional iterative learning is needed.

[0044] In some embodiments, physical elements of a thermodynamic chip may be used to physically model evolution according to Langevin dynamics. For example, in some embodiments, a thermodynamic chip includes a substrate comprising oscillators implemented using superconducting flux elements. The oscillators may be mapped to neurons (visible or hidden) that “evolve” according to Langevin dynamics. For example, the oscillators of the thermodynamic chip may be initialized in a particular configuration and allowed to thermodynamically evolve. As the oscillators “evolve” degrees of freedom of the oscillators may be sampled. Values of these sampled degrees of freedom may represent, for example, vector values for neurons or synapses that evolve according to Langevin dynamics. For example, algorithms that use stochastic gradient optimization and require sampling during training, such as those proposed by Welling and Teh, and/or other algorithms, such as natural gradient descent, mirror descent, etc. may be implemented using a thermodynamic chip. In some embodiments, a thermodynamic chip may enable such algorithms to be implemented directly by sampling the neurons and/or synapses (e.g., degrees of freedom of the oscillators of the substrate of the thermodynamic chip) without having to calculate statistics to determine probabilities. As another example, thermodynamic chips may be used to perform autocomplete tasks, such as those that use Hopfield networks, which may be implemented using the Welling and Teh algorithm. For example, visible neurons may be arranged in a fully connected graph (such as a Hopfield network, etc.), and the values of the auto complete task may be learned using the Welling and Teh algorithm. In some embodiments, a relay gadget may be used to sample visible neurons of a first energy-based model (EBM) and provide the sampled thermodynamic information as an input to another energy-based model (EBM).

[0045] In some embodiments, a thermodynamic chip includes superconducting flux elements arranged in a substrate, wherein the thermodynamic chip is configured to modify magnetic fields that couple respective ones of the oscillators with other ones of the oscillators. In some embodiments, non-linear (e.g., anharmonic) oscillators are used that have dual-well potentials. These dual-well oscillators may be mapped to neurons of a given energy-based model that the thermodynamic chip is being used to implement. Also, in some embodiments, at least some of the oscillators may be harmonic oscillators with single-well potentials. In some embodiments, oscillators may be implemented using superconducting flux elements with varying amounts of non-linearity. In some embodiments, an oscillator may have a single well potential, a dual-well potential, or a potential

somewhere in a range between a single-well potential and a dual-well potential. In some embodiments, visible neurons may be mapped to oscillators having a single well potential, a dual-well potential, or a potential somewhere in a range between a single-well potential and a dual-well potential.
[0046] In some embodiments, oscillators of a thermodynamic chip may also be used to represent values of weights and biases of the energy-based model. Thus, weights and biases that describe relationships between neurons may also be represented as dynamical degrees of freedom, e.g., using oscillators of the thermodynamic chip (e.g., synapse oscillators).
[0047] In some embodiments, parameters of an energy-based model or other learning algorithm may be learned through evolution of the oscillators of a thermodynamic chip.
[0048] As mentioned above, in some embodiments, the weights and biases of an energy-based model may be dynamical degrees of freedom (e.g., oscillators of a thermodynamic chip), in addition to neurons (hidden or visible) being dynamic degrees of freedom (e.g., represented by other oscillators of the thermodynamic chip). In such configurations, gradients needed for learning algorithms can be obtained by performing sampling of the synapse oscillators, such as position samples or momentum samples. For example, measurements of the synapse oscillators (position or momentum) performed on a time scale proportional to a thermalization time of the synapse oscillators, or on shorter time scales than the thermalization times of the synapse oscillators, can be used to compute time-averaged gradients in addition to space averaged gradients. In some embodiments, the variance of the time averaged or space averaged gradient (determined using synapse oscillator measurements) scales as 1/t where t is the total measurement time. These gradients can be used to calculate new weights and bias values that may be used as synapse values in an updated version of the energy-based model. The process of making measurements and determining updated weights and biases may be repeated multiple times until a learning threshold for the energy-based model has been reached.
[0049] For example, there are various learning algorithms where one must use both positive and negative phase terms to perform parameter updates. For instance, in the implementation by Welling and Teh the parameters are updated as follows:

[00001] $$_{t+1} = {}_t + \tfrac{{}_t}{2}(-\nabla_{t}\, {}_p({}_t) - N\tfrac{1}{n}\ .\mathrm{Math.}_{i=1}^{n}\ \nabla_{t}\,({}_t, x_{t_i}) - \mathbb{E}_{x \sim p_{t}(x)}[\nabla_{t}\,({}_t, x)]) + {}_t$$

where ε.sub.p (θ.sub.t) is some prior potential and the probability distribution for an energy-based model (EBM) with parameters θ.sub.t given by pθ.sub.t (x)=e.sup.−ε(θ.sup.t.sup.,x)/Z, where Z is a partition function. In the above equation, the first gradient term, where the visible nodes are clamped to the data will be referred to as the positive phase term. The second gradient term, where the visible nodes are sampled from x˜pθ.sub.t(x) will be referred to as the negative phase term (e.g., where the visible nodes are unclamped). When hidden neurons are present, the parameter update rule is given by:

[00002] $$_{t+1} = {}_t + \tfrac{{}_t}{2}(-\nabla_{t}\, {}_p({}_t) - N\tfrac{1}{n}\ .\mathrm{Math.}_{i=1}^{n}\ \mathbb{E}_{z \sim p_{t}(z \mid x_{t_i})}[\nabla_{t}\,({}_t, x_{t_i}, z)] - \mathbb{E}_{(x,z) \sim p_{t}(x,z)}[\nabla_{t}\,({}_t, x, z)]) + {}_t$$

[0050] A selection of experts gadget as described herein uses parameter updates θ.sub.j,t+1 (for expert energy-based model parameters) and η.sub.t+1 (for SoftMax parameters) which are based on the general parameter update rule above. The parameter updates θ.sub.j,t+1 and η.sub.t+1 are further described with regard to FIG. **1**D.
[0051] For a neuro-thermodynamic processor, which includes visible neurons coupled via weights and biases that are also represented by degrees of freedom (e.g., synapse oscillators), the dynamics of the system for a three-body coupling between the synapse oscillators and the neuron oscillators (visible or hidden) are described by the following Hamiltonian:
[00003]

$$H_{\mathrm{total}} = \ .\mathrm{Math.}_{j \in {}_{\mathrm{vis}}}\ (\frac{p_{n_j}^2}{2m_{n_j}^{(v)}} + E_L^{(v)}({}_{n_j} - \tilde{}_L^{(v)})^2 + E_{J0}^{(v)}\cos(\tilde{}_{DC}^{(v)}/2)(1-\cos({}_{n_j}))) + \ .\mathrm{Math.}_{k,l \in}\ (\frac{p_{s_j}^2}{2m_{s_{k,l}}^{(w)}} + E_L^{(w)}({}_{s_{k,l}} - \tilde{}_L^{(w)})^2 + E_{J0}^{(w)}\cos(\tilde{}_{DC}^{(w)}/2)(1-\cos({}_{s_{k,l}}))) + \ .\mathrm{M}_{j \in}$$

[0052] Note that the above Hamiltonian uses a representation of couplings between neuron oscillators and synapse oscillators given by the terms proportional to alpha and beta. However, in some embodiments, a Hamiltonian with more general terms may be used. The above Hamiltonian is given as an example of an energy-based model, but others may be used within the scope of the present disclosure.
[0053] Broadly speaking, classes of algorithms that may benefit from implementation using a thermodynamic chip include those algorithms that involve probabilistic inference. Such probabilistic inferences (which otherwise would be performed using a CPU or GPU) may instead be delegated to the thermodynamic chip for a faster and more energy efficient implementation. At a physical level, the thermodynamic chip harnesses electron fluctuations in superconductors coupled in flux loops to model Langevin dynamics. In some embodiments, architectures such as those described herein may resemble a partial self-learning architecture, wherein classical computing device(s) (e.g., a FPGA, ASIC, etc.) may be relied upon only to perform simple tasks such as summing measured values and performing other non-compute intensive operations in order to implement a learning algorithm.
[0054] Note that in some embodiments, electro-magnetic or mechanical (or other suitable) oscillators may be used. A thermodynamic chip may implement neuro-thermodynamic computing and therefore may be said to be neuromorphic. For example, the neurons implemented using the oscillators of the thermodynamic chip may function as neurons of a neural network that has been implemented directly in hardware. Also, the thermodynamic chip is “thermodynamic” because the chip may be operated in the thermodynamic regime slightly above 0 Kelvin, wherein thermodynamic effects cannot be ignored. For example, some thermodynamic chips may be operated within the milli-Kelvin range, and/or at **2**, **3**, **4**, etc. degrees Kelvin. The term thermodynamic chip also indicates that the thermal equilibrium dynamics of the neurons are used to perform computations. In some embodiments, temperatures less than 15 Kelvin may be used. Though other temperatures ranges are also contemplated. This also, in some contexts, may be referred to as analog stochastic computing. In some embodiments, the temperature regime and/or oscillation frequencies used to implement the thermodynamic chip may be engineered to achieve certain statistical results. For example, the temperature, friction (e.g., damping) and/or oscillation frequency as well as masses, may be controlled variables that ensure the oscillators evolve according to a given dynamical model, such as Langevin dynamics. In some embodiments, temperature may be adjusted to control a level of noise introduced into the evolution of the neurons. As yet another example, a thermodynamic chip may be used to model energy models that require a Boltzmann distribution. Also, a thermodynamic chip may be used to solve variational algorithms and perform learning tasks and operations.
[0055] FIG. **1**A is a high-level block diagram of a thermodynamic chip arrangement for a selection of experts gadget, according to some embodiments.
[0056] A thermodynamic chip **122** may implement a selection of experts gadget **132** using thermodynamic oscillators. Thermodynamic oscillators may also be used to implement relay gadgets, which may store and transmit data from one energy-based model to another. Input data may be stored using the input data oscillator(s) **120**, which may be initialized to the input data using a traditional computing system, may be output oscillators of another energy-based model, or may be relay gadgets which received data from output oscillators of another energy-based model. The input data oscillator(s) **120** may be configured to connect to a SoftMax gadget **106** and a multiplication gadget **102**. The SoftMax gadget **106** is further described with respect to FIGS. **3** and **4**.
[0057] The SoftMax gadget **106** may have a set of input/output oscillators instead of discreet sets of input oscillators and output oscillators. Input data oscillator(s) **120** may connect to the input/output oscillators of the SoftMax gadget **106**. In some embodiments the input/output oscillators of the SoftMax gadget **106** may be the SoftMax data sample receiver oscillators **104**, and in some embodiments the SoftMax data sample receiver oscillators **104** may be relay gadgets, such as relay oscillators. Relay gadgets are further described with respect to FIGS. **7**A-**12**.
[0058] The multiplication gadget **102** may accept input data from the input data oscillator(s) **120** and SoftMax data from the SoftMax data sample receiver oscillators **104**. The multiplication gadget **102** may modify the input data based on the SoftMax data. Dimensions of a SoftMax vector may correspond to different expert energy-based models (i.e., first energy based model **110**, second energy based model **114**, and nth energy based model **118**). Data samples of a SoftMax gadget **106** may be one-hot encoded vectors, or a weighted average of one-hot encoded vectors. The multiplication

gadget **102** may modify the input data with a function which multiplies the input data by the SoftMax data, as explained with respect to FIG. **1**E. Output oscillators of a multiplication gadget **102** (and relay oscillators connected to the output oscillators) may correspond to expert energy-based models. For example, first data sample receiver oscillators **108** may accept modified input data corresponding to first energy based model **110**, second data sample receiver oscillators **112** may accept modified input data corresponding to second energy based model **114**, and nth data sample receiver oscillators **116** may accept modified input data corresponding to nth energy based model **118**.

[0059] The expert energy based models (i.e., first energy based model **110**, second energy based model **114**, and nth energy based model **118** may use the modified input data to generate output data, which the expert energy based models may send to output relay oscillators **130**. The output relay oscillators **130** may store the output of the selected expert, and may pass the output along as input to oscillators of another thermodynamic gadget, or be read out by a standard computing device.

[0060] Input data oscillator(s) **120** may be a set of oscillators, and each oscillator may represent a dimension of a vector. Collectively, input data oscillator(s) **120** may represent the vector, which may be a data sample generated by another energy-based model. Each of the input oscillator sets for the expert EBMs (i.e., each of first data sample receiver oscillators **108**, second data sample receiver oscillators **112**, and nth data sample receiver oscillators **116**) may be a set of oscillators with a number of oscillators equal to the number of input data oscillator(s) **120**. The SoftMax gadget **106** input/output oscillators (and SoftMax data sample receiver oscillators **104**) may be a set of oscillators with a number of oscillators equal to the number of expert energy-based models, i.e., a set of n oscillators. A SoftMax data sample may be a one-hot encoded vector with n dimensions, wherein the single dimension which has a value of 1 instead of 0 indicates the expert energy based model which is to process the input data.

[0061] FIG. **1**B is a block diagram of a set of SoftMax output oscillators in a particular one-hot vector configuration which selects a particular energy-based model, according to some embodiments.

[0062] The configuration of the SoftMax output oscillators **106**A shown in FIG. **1**B can be represented by the one-hot encoded vector <1, 0, . . . , 0>. The oscillator corresponding to the first energy based model **110** has an approximate value of 1, encoded in the potential degree of freedom of the oscillator. The oscillator corresponding to the second energy based model **114** and the oscillator corresponding to the nth energy based model **118** both have a value of approximately 0, encoded in the potential degree of freedom of the oscillators. Unillustrated SoftMax output oscillators also have an approximate value of 0. The one-hot encoded vector <1,0, . . . ,0> selects the first expert EBM out of N expert EBMs.

[0063] FIG. **1**C is the high-level block diagram of the thermodynamic chip arrangement with the SoftMax output oscillators in the particular one-hot vector configuration, according to some embodiments.

[0064] Because the SoftMax output oscillators **106**A have the configuration <1,0, . . . ,0>, the first energy based model **110** is selected. The multiplication gadget multiplies the SoftMax vector supplied by SoftMax output oscillators **106**A with the input data vector supplied by the input data oscillators(s) **120**. The data sample receive oscillators (i.e., first data sample receiver oscillator **108**, second data sample receiver oscillator **112**, and nth data sample receiver oscillator **116**) receive modified input data based on the SoftMax data sample. As a result, the first data sample receiver oscillator **108** receives a vector that is close to the input data vector, and the other data sample receiver oscillators (i.e., second data sample receiver oscillator **112** and nth data sample receiver oscillator **116**) receive vectors of the same dimension as the input data vector with values close to 0 at each dimension. The first energy based model **110** may be able to process the modified input data received by the first data sample receiver oscillator **108** and generate output to transmit to output relay oscillators **130**A.

[0065] FIG. **1**D is a block diagram of a configuration for an overall selection of experts gadget which produces results using a set of energy-based models, according to some embodiments.

[0066] The equation shown for the potential of the selection of experts gadget **132** includes terms representing the potential of input oscillators, the potential of data sample receiver oscillators, the potential of the SoftMax gadget, the potential of the multiplication gadget, and the potential of expert energy-based models.

[00004]

$$V(\ _x,\ _s,\ _z,\ ,\ ) = \tfrac{1}{2}m_x\ {}_x^2\ .\text{Math}._l\ (\ _{x_l} - x_l)^2 + \tfrac{1}{2}m_z\ {}_z^2\ .\text{Math}._l\ {}_{z_l}^2 + V_{s_1}(\ _s,\ ) + V_{s_2}(\ _s) + V_c(\ _s,\ _z,\ _x) + .\text{Math}._{j=1}^{N}\ _j(\ _{z_j},\ _j)$$

[0067] The term ½ m.sub.xω.sub.x.sup.2Σ.sub.l(ϕ.sub.x.sup.l−x.sub.l).sup.2 corresponds to the overall potential of the input data oscillators (ϕ.sub.x, with an individual oscillator represented by ϕ.sub.x.sub.l). m.sub.x represents the mass of the input data oscillators, and ω.sub.x.sup.2 represents the frequency squared of the input data oscillators. The input data oscillators collectively represent an input data vector, with K dimensions where l∈{1, . . . , K}. The input data vector may be represented as x with individual dimensions represented as x.sub.l when the input data is clamped to the input data oscillators.

[0068] The term ½ m.sub.zω.sub.z.sup.2Σ.sub.lϕ.sub.z.sub.l.sup.2 corresponds to the overall potential of the data sample receiver oscillators (ϕ.sub.z, with an individual oscillator represented by ϕ.sub.z.sub.l). m.sub.z represents the mass of the data sample receiver oscillators, and ω.sub.z.sup.2 represents the frequency squared of the data sample receiver oscillators. The data sample receiver oscillators collectively represent a modified input vector, with K dimensions where l∈{1, . . . , K}. A single set of data sample receiver oscillators may have a non-zero potential at any given time as the modified input data may be a matrix including vectors which are close to zero vectors and one vector which is close to the input vector.

[0069] The terms V.sub.s.sub.1 (ϕ.sub.s, η) and V.sub.s.sub.2 (ϕ.sub.s) correspond to the SoftMax gadget. The term V.sub.s.sub.1 (ϕ.sub.s, η) is further described by the equation V.sub.s.sub.1 (ϕ.sub.s, η)=λ.sub.s Σ.sub.j=1.sup.N ϕ.sub.s.sub.jη.sub.j where λ.sub.s is a coupling parameter for the SoftMax gadget, ϕ.sub.s represents the oscillators of the SoftMax gadget, ϕ.sub.s.sub.j represents an individual oscillator of the SoftMax gadget which corresponds to a j expert energy-based model of N total expert energy-based models, where j∈{1, . . . , N}, where η represents the learnable SoftMax parameters, and where η.sub.j represents an individual learnable SoftMax parameter. The term V.sub.s.sub.2 (ϕ.sub.s) is further described by the equation V.sub.s.sub.2 (ϕ.sub.s)=A.sub.1Σ.sub.j=1.sup.Nϕ.sub.s.sub.j.sup.2(1−ϕ.sub.s.sub.j).sup.2+A.sub.2 (Σ.sub.j=1.sup.Nϕ.sub.s.sub.j−1), where A.sub.1 and A.sub.2 are large coupling terms which cause the ϕ.sub.s SoftMax oscillators to be one-hot encoded vectors as described further with respect to FIG. **4**.

[0070] The term V.sub.c (ϕ.sub.s, ϕ.sub.z, ϕ.sub.x) corresponds to the multiplication gadget and is further described in relation to FIG. **1**E. The term Σ.sub.j=1.sup.Nε.sub.j (ϕ.sub.z.sub.j, θ.sub.j) corresponds to the N expert energy-based models and is dependent on the potential energy of each of the N expert energy-based models. θ represents all the learnable expert energy-based model parameters, and θ.sub.j represents an individual set of learnable expert energy-based model parameters corresponding to an individual expert energy-based model.

[0071] The parameters for the SoftMax gadget (η) and the expert energy-based models (θ.sub.j) may be set during initialization or trained using positive and negative phase training over time t as described by the following equations for θ.sub.j,t+1 and η.sub.t+1, where ξ.sub.t˜N(0, ∈.sub.t):

[00005]

$$_{j,t+1} = \ _{j,t} - \tfrac{t}{2}N(\tfrac{1}{n}\ .\text{Math}._{i=1}^{n}\ \mathbb{E}_{\ _{z_j} \sim p(\ _s,\ _z | x_a)}[\nabla_{\ _{j,t}}\ _j(\ _z,\ _{j,t})] - \mathbb{E}_{\ _{z_j} \sim p(\ _s,\ _z,\ _x)}[\nabla_{\ _{j,t}}\ _j(\ _z,\ _{j,t})]) + \ _t \quad _{t+1} = \ _t - \tfrac{t}{2}N(\tfrac{1}{n}\ .\text{Math}._{i=1}^{n}\ \mathbb{E}_{\ _s \sim p(\ _s,\ _z}$$

[0072] The probability distribution

[00006] $p(\ _s,\ _z,\ _x) = \frac{e^{-\ (V(\ _x,\ _s,\ _z,\theta\eta))}}{Z(\ ,\ )}$

uses the partition function Z(θ, η)=∫dϕ.sub.sdϕ.sub.xdϕ.sub.ze.sup.−βV(ϕ.sup.x.sup.,ϕ.sup.s.sup., ϕ.sup.z.sup.,θ,η), where β is the thermodynamic beta. Due to V.sub.s.sub.1 and V.sub.s.sub.2 causing the SoftMax distribution to be one-hot encoded vectors, the partition function can be re-written as

[00007] $Z(\ ,\ ) \approx\ .\text{Math}._{j=1}^{N}\ e^{-\ [V_{s_1}(s_j = 1,\ ) + V_{s_2}(s_j = 1)]} \int d\ _x\, e^{-\ \ _j(\ _x,\ _j)}\ .\text{Math}._{i \neq j}\ d\ _{z_j}\, e^{-\ \ _i(\ _{z_i},\ _i)},$

over all one-hot encoded vectors of size N, where ϕ.sub.s.sub.j=1 and ϕ.sub.s.sub.i=0 for all i≠j. Where the coupling parameter λ.sub.c in the multiplication gadget potential equation, which is described in further detail with respect to FIG. **1**E, is large such that ϕ.sub.z.sub.j≈ϕ.sub.x, all integrals of the partition function evaluate to functions which only depend on the parameters θ=(θ.sub.1, . . . , θ.sub.N), and the partition function can be further re-written as

[00008] $Z(\ , \ ) \approx$ .Math.$_{j=1}^{N}$ $e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]}$ .Math.$_{i=1}$ $Z(\ _i)$.

[0073] The parameter updates for θ.sub.j,t+1 and η.sub.t+1 use gradients (∇.sub.θ.sub.j,t) for the positive phase term, where input oscillators are clamped to input data, i.e., ϕ.sub.x=x.sub.t.sub.a where a is an index for a data element (x) batch, and the negative phase term where input oscillators are unclamped. The gradients may be obtained by measurement of the position or momentum of the θ and η oscillators. The θ and η parameters may be repeatedly initialized to θ.sub.t and η.sub.t values, and the average gradients may be used to determine the parameter updates for θ.sub.j,t+1 and η.sub.t+1 in the equations above.

[0074] A value for V (ϕ.sub.x, ϕ.sub.s, ϕ.sub.z, θ, η) may be obtained by accounting for the potentials used to initialize the oscillators θ and η with the equation

[00009] $V_{\text{total}}(\ _x, \ _s, \ _z, \ , \ ) = V(\ _x, \ _s, \ _z, \ , \ ) + \frac{1}{2}m \ ^2$ .Math.$_{j=1}^{N}$ $\ _j^2 + \frac{1}{2}m \ ^2$ .Math.$_{j=1}^{N}$ $\ _j$ .Math. $\ _j$,

assuming that the η.sub.j oscillators all have the same mass and frequency as each other and the θ.sub.j oscillators all have the same mass and frequency as each other. Additionally, the parameters θ and η as oscillators may have a longer thermodynamic equilibrium time than ϕ.sub.x, ϕ.sub.s, and ϕ.sub.z oscillators.

[0075] The thermal equilibrium time may be computing using the Fokker-Plank equation and computing the smallest eigenvalue. Particularly, where ϕ.sub.x oscillators have momentum π.sub.x and ψ.sub.0(ϕ.sub.x.sub., π.sub.x.sub.) is the thermal equilibrium distribution of the system, the thermal equilibrium time (t.sub.therm) can be found using the probability distribution

[00010] $p(\ _x, \ _x, t) \approx p_0 \ _0(\ _x, \ _x) + p_1 \ _0(\ _x, \ _x) e^{-\frac{t}{t_{\text{therm}}}}$.

Generally, larger mass times frequency squared of a system may result in longer thermal equilibrium time. Parameter oscillators (such as θ and η and A.sub.1 and A.sub.2) may have larger mass times frequency squared than oscillators used for data (such as ϕ.sub.x, ϕ.sub.s, and ϕ.sub.z).

[0076] The probability distribution p(ϕ.sub.x|θ, η)=∫dϕ.sub.sdϕ.sub.zp(ϕ.sub.x, ϕ.sub.s, ϕ.sub.z|θ,η) may define the likelihood of selecting the jth expert energy-based model out of N expert energy-based models. Using the partition function

[00011] $Z(\ , \ ) \approx$ .Math.$_{j=1}^{N}$ $e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]}$ .Math.$_{i=1}$ $Z(\ _i)$

and the fact V.sub.s.sub.2 (ϕ.sub.s) causes the ϕ.sub.s samples to be one-hot encoded vectors, the probability distribution can be re-written as:

[00012] $p(\ _x | \ , \ ) \approx \frac{1}{Z(\ , \ )}$ .Math.$_{j=1}^{N}$ $e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]} \int d \ _{z_j} e^{-[V_c(\ _x, \ _{z_j}, s_j=1) + \ _j(\ _{z_j}, \ _j)]}$ .Math.$_{i \neq j}$ $Z(\ _i)$

[0077] Because ϕ.sub.s.sub.j=1 and ϕ.sub.s.sub.i=0 for all i≠j, the integral can be re-written as

[00013] $Z(\ _i) = \int d \ _{z_i} e^{- \ _i(\ _z, \ _i)}$.

For the V.sub.c (ϕ.sub.s, ϕ.sub.z, ϕ.sub.x) potential, where λ.sub.c is large and

[00014] $e^{-V_c(\ _x, \ _{z_j}, s_j=1)} \approx \ (\ _x - \ _{z_j})$.

The probability distribution can be re-written as:

[00015] $p(\ _x | \ , \ ) \approx \frac{1}{Z(\ , \ )}$ .Math.$_{j=1}^{N}$ $e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]} e^{- \ _j(\ _x, \ _j)}$ .Math.$_{i \neq j}$ $Z(\ _i)$

[0078] Because

[00016] $Z(\ , \ ) \approx$ .Math.$_{j=1}^{N}$ $e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]}$ .Math.$_{i=1}$ $Z(\ _i)$,

the probability distribution can be further re-written as:

[00017] $p(\ _x | \ , \ ) \approx$ .Math.$_{j=1}^{N}$ $\frac{e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]}}{\text{.Math.}_{j=1}^{N} e^{-[V_{s_1}(s_j=1, \ )+V_{s_2}(s_j=1)]}} \frac{e^{- \ _j(\ _x, \ _j)}}{Z(\ _j)}$

and simplified to:

[00018] $p(\ _x | \ , \ ) \approx$ .Math.$_{j=1}^{N}$ $p(s_j=1 | \ )$ .Math.$_{i \neq j}$ $p(s_j=0 | \ ) \frac{e^{- \ _j(\ _x, \ _j)}}{Z(\ _j)}$

[0079] FIG. **1**E is a block diagram of a configuration for a multiplication gadget which accepts an input vector and results from a gating model such as a SoftMax gadget in order to select an energy-based model from a set of energy-based models to process the input vector, according to some embodiments.

[0080] The equation shown for the potential of the multiplication gadget **102** causes the data receiver oscillators to evolve to take on the input data vector where the SoftMax value is 1 for the associated expert energy-based model, and causes the data receiver oscillators to not take on the input data vector where the SoftMax value is 0 for the associated expert energy-based model.

[00019] $V_c(\ _s, \ _z, \ _x) = \ _c$ .Math.$_{j=1}^{N}$ $\ _{s_j}(\ _x - \ _{z_j})^2$

[0081] λ.sub.c is a coupling parameter for the multiplication gadget **102**. Other variables have the same meaning as discussed with respect to FIG. **1**D. The engineered potential V.sub.c (ϕ.sub.s, ϕ.sub.z, ϕ.sub.x) may cause the multiplication gadget to perform a multiplication function when λ.sub.c is large relative to the expectation values of the oscillators ϕ.sub.s, ϕ.sub.z, and ϕ.sub.x.

[0082] The result of the engineered potential V.sub.c (ϕ.sub.s, ϕ.sub.z, ϕ.sub.x) is that the input data vector stored in the input data oscillators (ϕ.sub.x) is multiplied by the SoftMax data sample (ϕ.sub.s), i.e., a one-hot encoded vector, and the resulting modified input data is transferred to the data receiver oscillators (ϕ.sub.z). Because the SoftMax data sample is a one-hot encoded vector, one set of data receiver oscillators (ϕ.sub.z.sub.j) receives the input data vector and the other sets of data receiver oscillators do not receive data. The set of data receiver oscillators (ϕ.sub.z.sub.j) which receives the input data vector is associated with the expert energy-based model which was selected by the SoftMax data sample.

[0083] As an example, the potential for the multiplication gadget **102** may implement the multiplication function using the SoftMax data sample [1, 0, . . . , 0] and input data [first value, second value, . . . , nth value] to generate the modified input data

[00020] $\begin{bmatrix} \text{firstvalue} & \text{secondvalue} & \text{.Math.} & \text{nthvalue} \\ 0 & 0 & \text{.Math.} & 0 \\ \text{.Math.} & \text{.Math.} & \text{.Math.} & \text{.Math.} \\ 0 & 0 & \text{.Math.} & 0 \end{bmatrix}$

where a set of data receiver oscillators (ϕ.sub.z.sub.j) represents a row, and an individual oscillator of the set of data receiver oscillators represents a column in the row. The data sample receiver oscillators may collectively represent the modified input data. The rows of the modified input data which contain zero values may be referred to as zero vectors. The row of the modified input data which contains a vector which is or is similar to the input data vector may be referred to as a non-zero vector. The row of the modified input data which contains the non-zero vector corresponds to the expert energy-based model which the one-hot encoded vector SoftMax data sample corresponds to.

[0084] FIG. **2**A is a block diagram of oscillators which are storing input data and a SoftMax data sample and oscillators of a multiplication gadget at a first time (T.sub.1) such as prior to couplings between the oscillators being established, according to some embodiments.

[0085] Oscillators may not have couplings established at all times, for example, at T.sub.1 the input data oscillator(s) **120** and the SoftMax data sample receiver oscillators **104** may not have couplings established with input oscillators of the example multiplication gadget **102**. The input data oscillator(s) **120** and the SoftMax data sample receiver oscillators **104** may have completed interactions with the SoftMax gadget at T.sub.1 such that the SoftMax data sample receiver oscillators **104** have a SoftMax data sample which the SoftMax gadget generated based on the input data of the input data oscillator(s) **120**. The function of the SoftMax gadget is described further with respect to FIG. **3**. In some embodiments, the SoftMax data sample receiver oscillators **104** may be relay oscillators or relay gadgets. In some embodiments, the SoftMax data sample receiver oscillators **104** may be input/output oscillators of the SoftMax gadget.

[0086] FIG. **2**B is a block diagram of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a second time (T.sub.2), such as after couplings between the oscillators are established and while thermodynamic evolution is occurring, according to some embodiments.

[0087] Couplings **200** may be established between the SoftMax data sample receiver oscillators **104** and the input oscillators of the multiplication gadget **102**. Couplings.sub.I **200** may also be established between the input data oscillators **120** and the input oscillators of the multiplication gadget **102**. The established couplings may enable the multiplication gadget **102** to use the SoftMax data sample and input data according to perform the multiplication function established by the engineered potential explained in relation to FIG. **1**E:

[00021]$V_C({}_{s}, {}_{z}, {}_{x}) = {}_{c}$ .Math.$_{j=1}^{N}$ ${}_{s_j}({}_{x} - {}_{z_j})^2$

[0088] λ.sub.c may relate to properties of couplings within the multiplication gadget **102**.

[0089] FIG. **2**C is a block diagram of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a third time (T.sub.3), such as after thermodynamic equilibrium is reached and the multiplication gadget has a data sample representing an expectation value vector, according to some embodiments.

[0090] The output **204** of the multiplication gadget **102** may be based on the one hot encoded vector **202** stored by the SoftMax data sample receiver oscillators **104**. The multiplication gadget **102** may generate as output **204** modified input data comprising a lack of data to be processed by expert energy-based models not selected by the one hot encoded vector **202**, and a vector similar to the input data vector stored by the input data oscillator(s) **120** to be processed by the expert energy-based model selected by the one hot encoded vector **202**.

[0091] FIG. **2**D is a block diagram of oscillators of the oscillators which are storing the input data and the SoftMax data sample and oscillators of the multiplication gadget at a fourth time (T.sub.4), such as such as after thermodynamic equilibrium is reached and a set of data sample receiver oscillators are able to receive the data sample representing the expectation value vector from the multiplication gadget, according to some embodiments.

[0092] The multiplication gadget **102** may transfer the output **204** illustrated in FIG. **2**C comprising modified input data to the data sample receiver oscillators (i.e., first data sample receiver oscillators **108**, second data sample receiver oscillators **112**, and nth data sample receiver oscillators) for the expert energy-based models (i.e., first energy based model **110**, second energy based model **114**, and nth energy based model **118**). The data sample receiver oscillators (i.e., first data sample receiver oscillators **108**, second data sample receiver oscillators **112**, and nth data sample receiver oscillators) may be relay oscillators or relay gadgets. In some embodiments, the data sample receiver oscillators (i.e., first data sample receiver oscillators **108**, second data sample receiver oscillators **112**, and nth data sample receiver oscillators) may be input oscillators which are included in the expert energy-based models (i.e., first energy based model **110**, second energy based model **114**, and nth energy based model **118**).

[0093] The expert energy-based models (i.e., first energy based model **110**, second energy based model **114**, and nth energy based model **118**) may use the modified input data obtained from the data sample receiver oscillators (i.e., first data sample receiver oscillators **108**, second data sample receiver oscillators **112**, and nth data sample receiver oscillators) to generate output for the overall selection of experts gadget **132**. The output for the overall selection of experts gadget **132** may be read out to a traditional computer, stored in relay oscillators or relay gadgets for later read out to a traditional computer or use in another energy-based model, or directly used as input in another energy-based model.

[0094] FIG. **3** illustrates an example coupling for an analog SoftMax gadget, wherein ancilla oscillators (ϕ.sub.a) are used to emulate an all-to-all coupling between input/output oscillators (ϕ.sub.b.sub.j) of the analog SoftMax gadget using at most degree-four connectivity, according to some embodiments.

[0095] In some embodiments, the input/output oscillators **300** of the analog SoftMax gadget **106** may be coupled to one another in an all-to-all coupling. The input/output oscillators **300** of the SoftMax gadget **106** may be relay oscillators, such as the oscillators **124**, **126**, and **128** illustrated in FIG. **1**B. In configurations with a large number of input/output oscillators, such an all-to-all configuration may be cumbersome to implement. Thus, in some embodiments a constructive all-to-all coupling may be used, wherein additional ancilla oscillators are configured in a modified tree-structure to achieve a constructive all-to-all coupling between input/output oscillators **300**. An example of a constructive all-to-all coupling is illustrated in FIG. **3**.

[0096] For example, a binary tree type of lattice (e.g. modified tree) may be used to create similar constraints on the input/output oscillators **300** as an all-to-all coupling. Note that in the engineered potential used to implement the SoftMax function, e.g., V(ϕ.sub.b.sub.1, . . . , ϕ.sub.b.sub.N)=A.sub.1Σ.sub.j=1.sup.Nϕ.sub.b.sub.j.sup.2 (ϕ.sub.b.sub.j−1).sup.2+A.sub.2 (Σ.sub.j=1.sup.Nϕ.sub.b.sub.j−1).sup.2, the A.sub.2 constant that is proportional to the second term (e.g., after the plus sign) requires an all-to-all coupling. This is because expanding this term will result in a term proportional to Π.sub.j=1.sup.Nϕ.sub.b.sub.j. However, in some embodiments, additional ancillary oscillators **302** as shown in FIG. **3**, may be added to reduce the degree of connectivity required between individual ones of the oscillators. For example, for a vector of dimension N each individual oscillator may only require connectivity to four other oscillators as opposed to connectivity to all N−1 other ones of the input/output oscillators **300**, as would be the case in a true all-to-all coupling.

[0097] For discussion purposes, the additional ancilla oscillators **302** can be considered to belong to a layer (l) and within each layer there are j ancilla oscillators **302**. The layer that includes the root node (shown as ϕ.sub.a.sub.1 and ϕ.sub.a.sub.2) in FIG. **3** may be referred to as layer **1**, e.g. l=1. The next higher layer may be layer **2**, e.g., l=2, and so on. A constraint is imposed on the ancilla oscillators **302** in the lowest layer (l=1) such that A.sub.2.sup.(1) (ϕ.sub.a.sub.1.sup.(1)−1) for some large coupling parameter A.sub.2.sup.(1). This imposes an energetic penalty if ϕ.sub.a.sub.1.sup.(1) deviates from a value of 1. In a layer l the position degree of freedom of two sibling nodes, such as ϕ.sub.a.sub.j.sup.(l) and ϕ.sub.a.sub.j+1.sup.(l), are further labeled with a subscript s to indicate they are sibling nodes. For example, they may be labeled as ϕ.sub.a.sub.j,s.sup.(l) and ϕ.sub.a.sub.j+1,s.sup.(l). The set of siblings for a given layer above the first layer is S.sup.(l). The position degree of freedom of a parent of a given set of sibling nodes is labeled as ϕ.sub.a.sub.j,p.sup.(l-1). Using this notation, the potential can be written with an energy constraint, as follows:

[00022]$V_{\text{Ancillas}} = A_2^{(1)}({}_{a_1}^{(1)} - 1)^2 +$ .Math.$_{l=2}^{L-1}$ .Math.$_{j \in {}^{(l)}}$ $A_2^{(l)}({}_{a_{j,s}}^{(l)} + {}_{a_{(j+1),s}}^{(l)} - {}_{a_{j,p}}^{(l-1)})^2 +$ .Math.$_{j \in {}^{(L)}}$ $A_2^{(L)}({}_{\mathrm{ab}_{j,s}}^{(L)} + {}_{b_{(j+1),s}}^{(L)} - {}_{a_{j,p}}^{(L-1)})^2$

[0098] The above equation uses the fact that at the bottom of the tree (e.g., the oscillators below layer l=1), the oscillators corresponding to the leaf nodes are the original ϕ.sub.b.sub.j oscillators (e.g., the input/output oscillators **300**). The above potential adds an energetic penalty in the root node for these leaf nodes (e.g., ϕ.sub.a.sub.1 and ϕ.sub.a.sub.2 as shown in FIG. **3**) if the root node is not one. Additionally, an energetic penalty is added if the children of all of the root nodes do not sum to ϕ.sub.a.sub.1.sup.(1), which should be 1. These conditions are added recursively until the leaf nodes are reached. The overall potential is then updated as:

[00023]$V({}_{b_1},$ .Math. $, {}_{b_N}) = A_1$ .Math.$_{j=1}^{N}$ ${}_{b_j}^{2}({}_{b_j} - 1)^2 + V_{\text{Ancillas}}$.

[0099] FIG. **4** illustrates graphs of potentials for a given oscillator of the analog SoftMax gadget, wherein the given oscillator has a dual-well

potential. FIG. **4** further illustrates how increasing the parameter A.sub.1 in an engineered potential for the analog SoftMax gadget causes the walls and intermediate barrier between the two wells of the dual-well potential to be more steep, such that the dual-well oscillator is more likely to evolve to a value of 0 or 1 as required by the engineered potential for the analog SoftMax gadget, according to some embodiments.
[0100] The respective input/output oscillators **300** of the analog SoftMax gadget **106** may be implemented using dual-well potential oscillators. Furthermore, selecting an appropriate value for A.sub.1 that is large creates an energetic penalty for values other than zero or one. This is illustrated in FIG. **4**, wherein increasing the value of A.sub.1 increases the well walls and the barrier between the wells, such that the minima of each of the wells is at zero or one.
[0101] FIG. **5** illustrates an example attention block of a machine learning model, that may be implemented in an analog manner using one or more thermodynamic chips, wherein an analog selection of experts gadget is used at least in part to implement an addition and normalization layer, according to some embodiments.
[0102] In some embodiments, an analog SoftMax gadget may be used as part of a thermodynamic implementation of a machine learning model. For example, in some embodiments a plurality of energy-based models may be used to implement functions of an attention block **502**. For example, EBMs may be used to thermodynamically implement an Addition and Normalization layer **504**, a Feed Forward layer **506**, another Addition and Normalization layer **508**, a Self-Attention layer **510**, and Input Embedding **512**. More specifically, an analog SoftMax gadget **106** may be one of a set of EBMs used to implement the Self-Attention layer **510** and a selection of experts gadget **102** may be used to implement an Addition and Normalization layer **508**.
[0103] FIG. **6** is a flowchart illustrating a process for implementing a selection of experts function using an analog selection of experts gadget, according to some embodiments.
[0104] At block **602**, a set of input/output oscillators of an analog SoftMax gadget (or relay oscillators) and input oscillators are coupled to a set of input oscillators of a multiplication gadget, output oscillators of which are coupled to input oscillators of a set of expert EBMs.
[0105] Then, at block **604**, the oscillators of the system (including the input-output oscillators and ancilla oscillators (if used) of the SoftMax gadget) are allowed to thermally evolve, e.g. to reach a thermal equilibrium. This evolution is performed based on an engineered potential for the analog SoftMax gadget which creates energetic penalties that drive the oscillators to thermodynamically evolve to a one-hot encoded vector state (e.g., one input/output oscillator having a position degree of freedom value of one, and all other input/output oscillators having a position degree of freedom value of zero).
[0106] The engineered potential for the multiplication gadget causes a vector represented by the input data to be multiplied by the one-hot encoded vector such that one expert EBM is selected to process the input data. For example, at block **606** (after the thermal evolution) the selected expert EBM arrives at an analog result of the expert EBM's function based on the input data as a result of being selected by the one-hot encoded vector output by the SoftMax gadget.
[0107] At block **608**, the output oscillators of the selected expert EBM are coupled to another EBM or other device that is to receive the result. This could be another EBM, relay oscillators, measurement, etc.
[0108] As another alternative, at block **610** the output oscillators of the selected expert EBM are coupled to relay gadgets, such as shown in FIG. **2**D, wherein the relay gadgets have any of the configurations shown in FIGS. **9-12**. The relay gadgets store respective expectation values of the output oscillators of the selected expert EBM.
[0109] FIG. **7**A is high-level diagram illustrating a first energy-based model (EBM) implemented using a thermodynamic chip, a second energy-based model (EBM) implemented using a thermodynamic chip, and a relay gadget implemented using a thermodynamic chip, wherein the relay gadget is configured to relay thermodynamic information between the first energy-based model (EBM) and the second energy-based model (EBM), according to some embodiments.
[0110] In some embodiments, a relay oscillator gadget, such as relay oscillator gadget **710**, receives thermodynamic information from an input source, such as oscillator **706**, and relays the thermodynamic information to an output destination, such as oscillator **708**. In some embodiments, the oscillator **706** may be an output oscillator **706** of a first energy-based model (EBM) **700** and the oscillator **708** may be an input oscillator **708** of a second energy-based model (EBM) **702**. In some embodiments, the thermodynamic information being relayed from the output oscillator **706** to the input oscillator **108** may be a position degree of freedom. As such, FIG. **7**A shows an output position degree of freedom (ϕ.sub.y) of the output oscillator **706** and an input position degree of freedom (ϕ.sub.x) of the input oscillator **708**, as well as a relay position degree of freedom (ϕ.sub.r) of the relay oscillator **718** and a bias position degree of freedom (ϕ.sub.b) of the bias oscillator **712**. Additionally, controller **714** is shown, which may be an on-chip controller. Controller **714** causes pulses to be emitted in a time dependent manner to orchestrate coupling of the relay oscillator **710** to the output oscillator **706**, coupling of the relay oscillator **710** to the bias oscillator **712**, adjustment of a mass or frequency of the relay oscillator **710**, and a coupling of the relay oscillator **710** to the input oscillator **708**. In some embodiments, the controller **714** may be pre-programmed to emit the relevant pulses and control signals in a time dependent sequence in order to execute a relay operation.
[0111] An example Hamiltonian of the coupled system shown in FIG. **7**A is given by:
[00024]
$$H_{\text{fan}} = \frac{{}^{2}_{r}}{2m_r(t)} + \frac{{}^{2}_{y}}{2m_y} + \frac{{}^{2}_{x}}{2m_x} + \frac{{}^{2}_{b}}{2m_b} + \tfrac{1}{2}m_r(t)\ {}^{2}_{r}(t)\ {}^{2}_{r} + \tfrac{1}{2}m_b\ {}^{2}_{b}\ {}^{2}_{b}\tfrac{1}{2}m_y\ {}^{2}_{y}(\ {}_{y} - y_e)^2 + \tfrac{1}{2}m_x\ {}^{2}_{x}\ {}^{2}_{x} + \ {}_{A}(t)(\ {}_{y} - \ {}_{r})^2 + \ {}_{B}(t)\ {}_{b}\ {}_{r} + \ {}_{X}(t)\ {}_{r}\ {}_{X}$$
[0112] Note that the terms in the Hamiltonian including the λ.sub.A, λ.sub.B, and λ.sub.X terms describe the coupling between the relay oscillators and the other three oscillators, e.g., the output oscillator **706**, the bias oscillator **712**, and the input oscillator **708**. Also, note that all three coupling terms are time dependent, based on the λ.sub.A, λ.sub.B, and λ.sub.X pulses controlled by controller **714**. Additionally, note that the mass (or the frequency) of the relay oscillator **710** is time dependent, where the mass (or frequency) of the relay oscillator is also controlled by controller **714**.
[0113] More particularly, the controller **714** emits pulses λ.sub.A to couple the position degree of freedom (ϕ.sub.y) of the output oscillator **706** to the position degree of freedom (ϕ.sub.r) of the relay oscillator **710**. This coupling may remain turned on for some time. Then, once the coupling between the position degree of freedom (ϕ.sub.y) of the output oscillator **706** and the position degree of freedom (ϕ.sub.r) of the relay oscillator **710** is turned off, the controller **714** causes pulses λ.sub.B to be emitted to couple the position degree of freedom (ϕ.sub.r) of the relay oscillator **710** to the position degree of freedom (ϕ.sub.b) of the bias oscillator **712**, and simultaneously emits control signals to cause the mass of the relay oscillator **710** to be increased (or alternatively emits control signals to cause the oscillation frequency of the relay oscillator **710** to be tuned, for example decreased). When coupled to the relay oscillator **710**, the bias position degree of freedom (ϕ.sub.b) of the bias oscillator **712** acts as a bias to the relay oscillator **710** and helps to ensure that the relay position degree of freedom (ϕ.sub.r) of the relay oscillator **710** maintains its equilibrium value (that it has acquired from the output oscillator **706**). After the relay oscillator **710** has reached an appropriately large mass (or tuned frequency), the controller **714** causes pulses λ.sub.X to be emitted to couple the position degree of freedom (ϕ.sub.r) of the relay oscillator **710** (having the increased mass or tuned frequency) to the position degree of freedom (ϕ.sub.X) of the input oscillator **708**. Also, in some embodiments, the controller **714** may cause pulses λ.sub.X and pulses λ.sub.B to be emitted at the same time, such that the relay oscillator **710** is coupled to the bias oscillator **712** simultaneously with being coupled to the input oscillator **708**. Note that in the illustration shown in FIG. **7**A either of EBMs **700** or **702** may be an analog SoftMax gadget **106**, that is to say the input to the relay oscillator may come from the analog SoftMax gadget **106** or the destination of the information being relayed may be the SoftMax gadget **106**. FIG. **7**A is illustrating a more general case for the relay gadget where the inputs and outputs are general EBMs, but it should be understood that the analog SoftMax gadget and analog multiplication gadget are particular implementations of EBMs having an engineered potential that implements a function.
[0114] In some embodiments the following pulse shapes may be used for λ.sub.A, λ.sub.B, and λ.sub.X. Though in some embodiments, other

suitable pulse shapes may be used.

[00025] $_A(t) = \ _A(\ (k_A(t-t_1)) - \ (k_A(t-t_2))) \ _B(t) = - \ _B \ (k_B(t-t_1^{(B)})) + \ _0^{(B)} \ _X(t) = \ _X \ (k_X(t-t_1^{(X)})) + \ _0^{(X)}$

where σ(t) is the sigmoid function:

[00026] $(t) = \frac{1}{1+e^{-t}}$.

[0115] In some embodiments, λ.sub.A, λ.sub.B, and λ.sub.X, as well as k.sub.A, k.sub.B, and k.sub.X may be tuned to improve results. Also, t.sub.1, t.sub.2, t.sub.1.sup.(B), and t.sub.1.sup.(X) may be tuned.

[0116] Without loss of generality, the position degree of freedom of the output oscillator **706** (ϕ.sub.y) is considered to have an equilibrium value (y.sub.e) (after energy-based model **700** has evolved for some time and reached a thermal equilibrium). Also, the position degree of freedom (ϕ.sub.y) of the output oscillator **706** is considered to have a potential given by ½m.sub.yω.sub.y.sup.2(ϕ.sub.y−y.sub.e).sup.2. It should be noted in practice that the output oscillator **706** may be coupled to various other oscillators of the first energy-based model **700** (as shown in FIG. **7**A) which would cause it to have the y.sub.e equilibrium value. Thus, to be more comprehensive, ½m.sub.yω.sub.y.sup.2(ϕ.sub.y−y.sub.e) may be replaced by a potential term that takes into account these couplings, such as ½ m.sub.y ω.sub.y.sup.2(ϕ.sub.y−y.sub.e).sup.2+Σ.sub.jλ.sub.Y.sup.(j)ϕ.sub.yϕ.sub.j or ½=m.sub.yω.sub.y.sup.2 (ϕ.sub.y−y.sub.e).sup.2+λ.sub.YΣ.sub.jλ.sub.Y.sup.(j) (ϕ.sub.y−ϕ.sub.j).sup.2, where the ϕ.sub.j degrees of freedom are degrees of freedom of other oscillators in the first energy-based model **700** that are coupled to the position degree of freedom (ϕ.sub.y) of the output oscillator **706**. However, this difference (or said another way, simplification) manifests itself in a slightly different value for the equilibrium value (y.sub.e), or depending on the couplings, may result in the same y.sub.e equilibrium value. But this simplification does not affect the equilibrium results of the relay oscillator **710**. A similar issue applies to the input oscillator **708**, which is also coupled to other oscillators of the second energy-based model **702**. Also, in some embodiments, multiple relay oscillators **710** may be coupled to multiple input oscillators (e.g. additional input oscillators in addition to input oscillator **708**). Note that the relay oscillator **710** and the relay gadget **704** impart the equilibrium value of the output oscillator to the input oscillator, such that the position degree of freedom (ϕ.sub.X) of the input oscillator **708** inherits the same equilibrium value as the position degree of freedom (ϕ.sub.y) of the output oscillator **706**, e.g. the position it had when first coupled to the relay oscillator **710** of the relay gadget **704**. As such, thermodynamic information is relayed from the output oscillator **706** to the input oscillator **708** while remaining in a thermodynamic state. For example, analog information is passed between the first energy-based model **700** and the second energy-based model **702** without requiring a measurement by a classical computing device. Further note, this is done in an analog way (as opposed to a digitization that would take place during readout and re-initialization).

[0117] For a system undergoing Langevin dynamics, the equation of motion of a given oscillator (k) is given by:

[00027] $\frac{d \ _k(t)}{dt} = \frac{\partial H_{fan}}{\partial \ _k} \frac{_k(t)}{dt} = - \ _k(t) - \frac{\partial H_{fan}}{\partial \ _k}\Big|_t + \sqrt{2m_k \ k_B T}\frac{dW_t}{dt}$

[0118] where ϕ denotes the position degree of freedom of the oscillator and It denotes the momentum degree of freedom of the oscillator. Using the Hamiltonian for the coupled system shown in FIG. **7**A (which is given further above) and the equations of motion for position and momentum given directly above, the equations of motions for the relay oscillator **710**, output oscillator **106**, the bias oscillator **712**, and the input oscillator **708**, are respectively given by:

[0119] Equation of motion for the relay oscillator:

[00028]

$m_r(t)\frac{d^2 \ _r}{dt^2} + \frac{dm_r(t)}{dt}\frac{d \ _r}{dt} + \ m_r(t)\frac{d \ _r}{dt} = -(-2 \ _A(t)(\ _y - \ _r) + \ _B(t) \ _b + \ _X(t) \ _x + m_r(t) \ _r^2 \ _r) + \sqrt{2m_r(t)k_B T}\frac{dW_t^{(r)}}{dt} \text{Or} \ m_r(t)\frac{d^2 \ _r}{dt^2} + \frac{dm_r(t)}{dt}\frac{d \ _r}{dt} + \ m_r(t)$

Depending on whether there is a linear or quadratic coupling.

[0120] Equation of motion for the output oscillator:

[00029]

$m_y\frac{d^2 \ _y}{dt^2} + \ m_y\frac{d \ _y}{dt} = -(\ _A(t) \ _y + m_y \ _y^2(\ _y - \ _c)) + \sqrt{2m_y k_B T}\frac{dW_c^{(y)}}{dt} \text{Or} \ m_y\frac{d^2 \ _y}{dt^2} + \ m_y\frac{d \ _y}{dt} = -(2 \ _A(t)(\ _y - \ _r) + m_y \ _y^2(\ _y - \ _c)) + \sqrt{2m_y k_B T}\frac{dW_t^{(y)}}{dt}$

Depending on whether there is a linear or quadratic coupling.

[0121] Equation of motion for the bias oscillator:

[00030]

$m_b\frac{d^2 \ _b}{dt^2} + \ m_b\frac{d \ _b}{dt} = -(\ _B(t) \ _r + m_b \ _b^2 \ _b) + \sqrt{2m_b k_B T}\frac{dW_t^{(b)}}{dt} \text{Or} \ m_b\frac{d^2 \ _b}{dt^2} + \ m_b\frac{d \ _b}{dt} = -(-2 \ _B(t)(\ _r - \ _b) + m_b \ _b^2 \ _b) + \sqrt{2m_b k_B T}\frac{dW_t^{(b)}}{dt}$

Depending on whether there is a linear or quadratic coupling.

[0122] Equation of motion for the input oscillator:

[00031]

$m_x\frac{d^2 \ _x}{dt^2} + \ m_x\frac{d \ _x}{dt} = -(\ _X(t) \ _r + m_x \ _x^2 \ _x) + \sqrt{2m_x k_B T}\frac{dW_t^{(x)}}{dt} \text{Or} \ m_x\frac{d^2 \ _x}{dt^2} + \ m_x\frac{d \ _x}{dt} = -(-2 \ _X(t)(\ _r - \ _x) + m_x \ _x^2 \ _x) + \sqrt{2m_x k_B T}\frac{dW_t^{(x)}}{dt}$

Depending on whether there is a linear or quadratic coupling.

[0123] Also, the time dependent mass of the relay oscillator **110** is given by:

[00032] $m_r(t) = m_f^{(r)} \ (k_r(t-t_r)) + m_r$.

[0124] FIG. **7**B is a high-level diagram similar to FIG. **7**A, wherein the relay gadget does not include a bias oscillator, according to some embodiments.

[0125] In some embodiments, such as when the relay oscillator is configured to have a controllable time-dependent mass, the use of a bias oscillator may be omitted. For example, if the product of mass times frequency squared of a first oscillator is much larger than the product of mass times frequency of a second oscillator (that is coupled to the first oscillator) the position degree of freedom of the first oscillator (having the larger value for the product of mass times frequency squared) may be treated as a constant. Thus, for embodiments, wherein the mass of the relay oscillator can be increased such that the product of mass times frequency squared of the relay oscillator is sufficiently large, it may not be necessary to further use a bias oscillator.

[0126] More particularly, consider two oscillators (oscillator a and oscillator b) with position degrees of freedom ϕ.sub.a and ϕ.sub.b. Suppose that ϕ.sub.b has equilibrium value b.sub.c. Assume ϕ.sub.b is a constant and consider the Hamiltonian:

[00033] $H_1 = \frac{1}{2}m_a \ _a^2 \ _a^2 + \ _a b_c$

In this case, the expectation value of φ.sub.a at thermal equilibrium is given by:

[00034] .Math. $_a$ .Math. $= \frac{\int ae^{-H_1}da}{\int e^{-H_1}da} = \frac{b_c}{m_a \ _a^2}$

Choosing λ=−m.sub.aω.sub.a.sup.2, it gives custom-characterϕ.sub.acustom-character=b.sub.c.

[0127] Also, considering the dynamics of ϕ.sub.b. The Hamiltonian is:

[00035] $H_2 = \frac{1}{2}m_a \ _a^2 \ _a^2 + \frac{1}{2}m_b \ _b^2(\ _b - b_c)^2 + \ _a \ _b$

Moreover, using H.sub.2, custom-characterϕ.sub.acustom-character is given by:

[00036] .Math. $_a$ .Math. $= \frac{\int ae^{-H_2}dadb}{\int e^{-H_2}dadb} = \frac{b_c}{m_a \ _a^2 - \frac{^2}{m_b \ _b^2}} = \frac{b_c}{1 - \frac{m_a \ _a^2}{m_b \ _b^2}}$

where λ is set such that λ=−m.sub.aω.sub.a.sup.2. Note that if m.sub.aω.sub.a.sup.2. <<m.sub.bω.sub.b.sup.2, then custom-characterϕ.sub.a custom-character≈b.sub.c. As such as long as the mass times frequency squared of the oscillator a having position degree of freedom ϕ.sub.a is

much less than the mass times frequency squared of the oscillator b having position degree of freedom ϕ.sub.b, the position degree of freedom ϕ.sub.b can be treated as a constant, with the constant being the thermal equilibrium value of ϕ.sub.b.

[0128] Said another way, if the product of mass times frequency squared of the relay oscillator **710** is increased to be sufficiently large, then the inherited equilibrium value acquired from the output oscillator **706** can be treated as a constant, while held by the relay oscillator **710**. Also, as long as the product of mass times frequency squared of the relay oscillator **710** is sufficiently large as compared to the corresponding value of mass times frequency squared of the input oscillator **708**, the position degree of freedom of the relay oscillator may be treated as a constant, such that it relays the held equilibrium value acquired from the output oscillator **706** of the first EBM **700** to the input oscillator **708** of the second EBM **702**.

[0129] Note that the relay oscillators used in the relay gadget configurations shown in FIGS. **9-12**, include bias oscillators. However, in some embodiments, similar configurations may be used that do not include bias oscillators. For example, relay oscillators as shown in FIG. **7**A or as shown in FIG. **7**B may be used to construct the relay gadgets shown in FIGS. **9-12**.

[0130] FIG. **8** is a high-level flowchart illustrating a process of relaying thermodynamic information between an output oscillator, such as of a SoftMax gadget, and an input oscillator, such as an input oscillator of a multiplication gadget, according to some embodiments.

[0131] At block **800** a relay oscillator is initialized, wherein the relay oscillator is positioned such that it has connectivity to an output oscillator, such as output oscillator **706** of energy-based model **700**, and has connectivity to an input oscillator, such as input oscillator **708** of energy-based model **702**. Additionally, a bias oscillator is initialized, wherein the bias oscillator has connectivity to the relay oscillator. For example, bias oscillator **712** may be initialized and is positioned in a way that it can be coupled to relay oscillator **710**.

[0132] At block **802**, the first energy-based model comprising the output oscillator, such as energy-based model **700** that includes output oscillator **706**, is enabled to undergo thermal evolution such that the energy-based model evolves according to Langevin dynamics. The evolution may be enabled to occur for an amount of time such that the first energy-based model reaches a thermal equilibrium. As an example, the first energy-based model may represent a trained model that is configured to perform inference, and at least some oscillators of the first energy-based model may be clamped to input data, wherein inference results are represented by other oscillators of the first energy-based model subsequent to the thermal evolution. For example, output oscillator **706** may represent the results of a computation performed by the energy-based model **700** that are to be relayed as input data to the second energy-based model **702**.

[0133] At block **804**, once the oscillators of the first energy-based model (e.g. energy-based model **700**) have reached thermal equilibrium, the controller **714** initiates pulses (e.g. λ.sub.A (t) pulses) to cause the output oscillator **706** to be coupled to the relay oscillator (e.g. relay oscillator **710**).

[0134] At block **806**, the controller **714** initiates additional pulses (e.g., λ.sub.B (t) pulses) that cause the relay oscillator to be coupled to the bias oscillator. Recall that initially the relay oscillator **710** may have a small mass and/or frequency combination, e.g., small relative to the product of mass times frequency squared of the output oscillator **706**. Because the relay oscillator has a small product of mass times frequency squared, the relay oscillator more readily takes on the position of the output oscillator (for example, as opposed to the relay oscillator pulling the output oscillator to take on the relay oscillator's position). However, due to the relatively small mass times frequency squared of the relay oscillator, if left alone the relay oscillator would quickly lose the recently inherited position, inherited from the output oscillator. To avoid this, the relay oscillator is coupled to the bias oscillator **712** at or near the same time as the relay oscillator is un-coupled from the output oscillator **706**. The relay oscillator may also be coupled to the bias oscillator at or near the same time it is coupled to the input oscillator **708**. Coupling the relay oscillator to the bias oscillator helps the relay oscillator to maintain the acquired thermal information (e.g., position degree of freedom, or, in some embodiments, momentum degree of freedom) the relay oscillator has acquired from the output oscillator. Also, while coupled to the bias oscillator and prior to being coupled to the input oscillator of the next EBM, a mass and/or frequency of the relay oscillator is adjusted.

[0135] For example, at block **808**, the controller **714** causes control signals to be emitted that cause the mass (or frequency) of the relay oscillator to be adjusted. The mass of the relay oscillator may be proportional to capacitance of a circuit used to implement the relay oscillator; a Cooper-pair box arrangement may be used to implement a time dependent capacitance in the circuit (e.g. where the capacitance corresponds to mass). In such embodiments, the controller **714** is configured to emit control signals to cause the Cooper-pair box to increase the capacitance of the relay oscillator circuit. However, in other embodiments, mass may be kept constant, but instead frequency of the relay oscillator may be adjustable as a result of a time-dependent flux element of a circuit used to implement the relay oscillator. For example, a current inducing flux element may be added to the relay oscillator circuit. In such embodiments, controller **714** may emit control signals that cause the flux of the relay oscillator to be tuned (where flux corresponds to frequency). In some embodiments blocks **806** and **808** are performed concurrently.

[0136] At block **810**, the controller **714** initiates another set of one or more pulses (e.g., λ.sub.X(t) pulses) to couple the relay oscillator to the input oscillator, such as input oscillator **708**. The bias oscillator **712** may remain coupled to the relay oscillator **710** when the relay oscillator **710** is coupled to the input oscillator **708**. Note that since the relay oscillator has had its mass (and/or frequency) adjusted prior to the coupling to the input oscillator, and since the relay oscillator remains coupled to the bias oscillator, the relay oscillator has a large value of the product of mass times frequency squared relative to the input oscillator and therefore causes the input oscillator to take on the position of the relay oscillator, which corresponds to the position of the output oscillator. In this way, the relay gadget **704** relays analog oscillator degree of freedom information (e.g. thermodynamic information) from the output oscillator to the input oscillator, without having to convert the thermodynamic information into classical form.

[0137] In some embodiments, a relay gadget, such as relay gadget **704**, may perform steps similar to those described in FIG. **8** in order to relay position degree of freedom thermodynamic information, momentum degree of freedom thermodynamic information, and/or force/acceleration degree of freedom thermodynamic information.

[0138] In some embodiments, a relay gadget, such as relay gadget **704** may be used to store thermodynamic information, for example in the relay oscillator **710**. Also, in some embodiments, multiple relay gadgets may be used to form a thermodynamic network between thermodynamic components. Also, in some embodiments, a relay gadget may be used to perform conditional sampling, such as Gibbs sampling.

[0139] FIG. **9** is a high-level diagram illustrating an output oscillator, an input oscillator, and a relay gadget, wherein the relay gadget comprises a group of relay oscillators and is configured to relay expectation values of thermodynamic information between the output oscillator and the input oscillator, according to some embodiments.

[0140] In some embodiments, it is desired to transfer an expectation value of one energy-based model (EBM) to another EBM, such as from an output of analog SoftMax gadget **106** to an input of another EBM. In some embodiments an instantaneous sample value may be transferred from an output oscillator of one EBM (such as from a given input/output oscillator **300** of analog SoftMax gadget **106**) to an input oscillator of another EBM. The instantaneous sample value of an output oscillator of a given EBM will follow a probability distribution associated with the potential well of the output oscillator and couplings of the output oscillator with the one or more oscillators belonging to the first EBM. An instantaneous sample value of the state of the output oscillator may be any possible value within the bounds of the potential well and respective couplings. In some instances, the instantaneous sample value of the output oscillator may be far off from the expectation value (e.g. due to thermodynamic fluctuations, anharmonic potentials, multiple well potentials, the coupling between the output oscillator with other oscillators belonging to a shared EBM, or a combination of factors). Furthermore, the output oscillator of an EBM may hop between wells of a potential, thus the expectation value may not be a probable outcome of an instantaneous sample of the output oscillator. To avoid these issues, in some embodiments expectation values may be stored instead of sample values and relayed as inputs to other EBMs.

[0141] In some embodiments, to enable an expectation value of an output of an EBM to be used as an input to a subsequent EBM in a fully analogue fashion (e.g. without the use of measurements), two or more relay oscillators may be used. In some embodiments, an expectation value is derivable from one or more sample values. In some embodiments, relay oscillators may be oscillators which may be arranged between the output of a given

EBM and the input of an additional EBM in such a way that their state may be configured to take on a sample value of the output oscillators of a given EBM. In some embodiments, sample values may be collected in such a way (e.g. spatial or temporal arrangement of relay oscillators as described below) that a close approximation of an expectation value of an output of a given EBM may be represented on one or more relay oscillators. Classical controllers may be used to turn the couplings on and off between the output oscillators and relay oscillators, between respective relay oscillators, as well as to make the masses and frequencies of the relay oscillators time dependent. Nevertheless, measurements may not be required, and the timing of the operations may be computed during a compilation step.

[0142] In some embodiments, a relay gadget may include a group of one or more relay oscillators and an additional relay oscillator. One or more relay oscillators of the group of relay oscillators may be coupled to an output oscillator of a first EBM. The one or more relay oscillators may be coupled in such a way that respective sample values of the output oscillator of the first EBM, wherein the output oscillator has progressed through thermodynamic evolution, may be stored on respective ones of the relay oscillators of the first group of one or more relay oscillators. An additional relay oscillator may be coupled to one or more of the relay oscillators, wherein the coupling enables the additional relay oscillator to take on an expectation value of the output oscillator, wherein the expectation value is derivable based at least in part on the sample values. In some embodiments, bias oscillators may be used. In some embodiments, bias oscillators may not be used. For simplicity, embodiments are given with bias oscillators, but it should be understood that is some embodiments bias oscillators may not be used for each relay oscillator of a relay gadget, however, that does not limit the embodiments to only one way or the other.

[0143] In some embodiments, thermodynamic information is relayed from a first energy-based model (EBM) **900** to a second energy-based model (EBM) **902** via relay gadget **904**. The thermodynamic information of EBM **900** is outputted via output oscillator **906** and inputted into input oscillator **908** via relay gadget **904**. The thermodynamic information may include, for example, samples of thermodynamic equilibrium of output oscillator **906**, or the expectation value of the output oscillator **906**. The expectation value is at least derivable based on samples values of the output oscillator **906**. Output oscillator **906** may be governed by a potential wherein the potential follows a single-well potential, double-well potential, multi-well potential, or any generic potential that may be engineered. The output oscillator **906** may also be coupled to other oscillators belonging to EBM **900**. More specifically, output oscillator **906** may be an input oscillator of analog selection of experts gadget **102**.

[0144] In some embodiments, an expectation value of one or more degrees of freedom of output oscillator **906** may be influenced by a potential of output oscillator **906** as well as couplings between output oscillator **906** and one or more oscillators belonging to first energy-based model **900**. Potentials governing the dynamics of the output oscillator **906** may have multiple wells. With generic arbitrary potentials (e.g. multiple wells) and coupling between output oscillator **906** and one or more oscillators belonging to first energy-based model **900**, the position degrees of freedom of the output oscillators can hop between wells. As described herein, a relay gadget provides a solution to approximate an expectation value of the output oscillator. For example, using an approximated expectation value in forwards and backwards propagation may provide better results than using a sample value, as the expectation value better represents the state of the oscillator whose degree of freedom value is being relayed to a second oscillator.

[0145] Relay gadget **904** comprises a group of relay oscillators **910** and an additional relay oscillator **912**. The group of relay oscillators **910** comprises one or more relay oscillators arranged with respective bias oscillators (e.g., relay oscillator **916** arranged with bias oscillator **918**). As described later, relay oscillators in oscillator group **910** may be configured and coupled in various ways (e.g. temporally and spatially) to transfer thermodynamic information. The additional relay oscillator **912** is connected to bias oscillator **920**. As discussed later, the additional relay oscillator **912** may be configured and coupled in various ways to transfer thermodynamic information. For example, the group of relay oscillators **910** transfers thermodynamic information to additional relay oscillator **912** via coupling **924**. Coupling **924** may be controlled by on-chip classical controller **914**.

[0146] Output oscillator **906** is coupled to the one or more relay oscillators of the group of relay oscillators **910** via on-chip classical controller **914**. On-chip classical controller **914** may send a pulse or a group of pulses to cause couplings between oscillators (e.g., coupling between output oscillator **906** and relay oscillator **916**) or relay oscillators like **916** and a bias oscillator like **918** via pulses **930**. Coupling is represented by coupling **922**, **924**, **926** and oscillators may be coupled or not coupled. When coupling is on, parameters of respective coupled oscillators affect the other oscillator it is coupled to. Couplings between oscillators within the group of relay oscillators **910** are not expressly shown in FIG. **9** to emphasize that the coupling may take different configurations (e.g. temporal or spatial configurations as detailed below). Nevertheless, on-chip classical controller **914** may cause a first set of one or more pulses to be emitted through controller connection **928**, wherein the first set of pulses couples one or more relay oscillators of the group of relay oscillators **910** to the output oscillator **906** (e.g., turn on coupling **922**). The on-chip classical controller **914** is further configured to cause a second set of one or more pulses to be emitted through path **932**, wherein the second set of pulses couples one or more relay oscillators of the group of relay oscillators **910** to the additional relay oscillator **912** (e.g., turn on coupling **924**). The on-chip classical controller **914** is further configured to cause a third set of one or more pulses (for example, set of pulses **938**) to be emitted, wherein the third set of pulses **938** couples the additional relay oscillator **912** to the input oscillator **108** (e.g., turn on coupling **926**).

[0147] In some embodiments, an additional relay oscillator **912** takes on an expectation value of an output oscillator **906** based at least in part on a coupling or couplings between a group of relay oscillators **910**, wherein respective relay oscillators of group **910** comprise respective sample values of the output oscillator **906**. The additional relay oscillator **912** may take on the expectation value of output oscillator **906** based at least on respective sample values taken on by respective relay oscillators. Furthermore, additional relay oscillator **912** may transfer the taken on expectation value to input oscillator **908** via controller **914** causing coupling **926** to turn on.

[0148] FIG. **10** is a high-level diagram illustrating a spatial analogue relay gadget, wherein respective ones of relay oscillators of a group of relay oscillators are configured to store respective sample values of an output oscillator, according to some embodiments.

[0149] In some embodiments, controller **914** sends a first set of one or more pulses wherein the first set of pulses causes output oscillator **906** of first energy-based model (EBM) **900** to be coupled to at least one or more relay oscillators {ϕ.sub.r.sub.1, ϕ.sub.r.sub.2, . . . ϕ.sub.r.sub.N}, in the group of relay oscillators **1010**. The group of relay oscillators **1010** comprises a plurality of relay oscillators, wherein respective relay oscillators {ϕ.sub.r.sub.1, ϕ.sub.r.sub.2, . . . ϕ.sub.r.sub.N}, are configured to store a sample of the output oscillator **906** based at least in part on respective couplings between the respective ones of the relay oscillators (e.g., **916**) of the group of relay oscillators **1010** and the output oscillator **906**. The on-chip classical controller **914** is further configured to cause another set of one or more pulses to be emitted, wherein the other set of pulses turns off the respective couplings between the output oscillator **906** and the respective ones of the relay oscillator of the group of relay oscillators **1010** at different times. This may allow different samples of the output oscillator **906** to be stored on the respective ones of the relay oscillators {ϕ.sub.r.sub.1, ϕ.sub.r.sub.2, . . . ϕ.sub.r.sub.N}.

[0150] On-chip classical controller **914** may be further configured to cause a second set of one or more pulses to be emitted, wherein the second set of pulses turns on the coupling between respective ones of the relay oscillators with sample values of the output oscillator **906** to an additional relay oscillator **1012**. The coupling is configured to transfer an approximation of the expectation value of output oscillator **906** based at least in part on the sample values stored on respective relay oscillators in the first group of relay oscillators **1010**. Once the additional relay oscillator **1012** is tuned to the expectation value of output oscillator **906**, controller **914** may cause a set of one or more pulses that may cause the additional relay oscillator **1012** to be coupled to input oscillator **908**. For ease of illustration a version that includes bias oscillators is shown. However, it should be understood that in some embodiments bias oscillators may be omitted.

[0151] FIG. **11** is a high-level diagram illustrating a temporal analogue relay gadget, wherein a group of relay oscillators comprises a single relay oscillator, according to some embodiments.

[0152] In some embodiments, the group of relay oscillators **910** comprises a single relay oscillator **1116**. The single relay oscillator **1116** is configured to store a sample of the output oscillator **906** based at least in part on the coupling between the single relay oscillator **1116** and the output oscillator **906**. The coupling between output oscillator **906** and single relay oscillator **1116** is caused by a first set of one or more pulses emitted from on-chip classical controller **914**. The on-chip classical controller **914** is configured to cause a second set of one or more pulses to be emitted, wherein the second set of pulses causes the single relay oscillator **1116** to be coupled to additional relay oscillator **1112**. The sequence of emitting the first set of pulses and then emitting the second set of pulses may be repeated numerous times. Each instance the sequence of the sequential sets of pulses is emitted, the position of additional relay oscillator **1112** is incrementally adjusted. Each adjustment may converge the additional relay oscillator **1112** to the expectation value of output oscillator **906**. For ease of illustration a version that includes bias oscillators is shown. However, it should be understood that in some embodiments bias oscillators may be omitted.

[0153] FIG. **12** is a high-level diagram illustrating a series analogue relay gadget, wherein a group of relay oscillators comprises a plurality of relay oscillators arranged in series, according to some embodiments.

[0154] FIG. **12** shows a drawing of a series analogue relay gadget **1204**. The group of relay oscillators **910** comprises a plurality of relay oscillators {ϕ.sub.r.sub.1, ϕ.sub.r.sub.2, . . . } (e.g. relay oscillator **1216**A, **1216**B, **1216**C) arranged one after another in series. Each relay oscillator has a product of mass and frequency squared. The first relay oscillator **1216**A, ϕ.sub.r.sub.1, has the smallest product of mass and frequency squared. The next relay oscillator **1216**B, ϕ.sub.r.sub.2, has a product of mass and frequency squared larger than the previous relay oscillator **1216**A, ϕ.sub.r.sub.1. This trend of increasing the product of mass and frequency squared continues for each subsequent relay oscillator in the group of relay oscillators **910**. As last in the chain of relay oscillators, the additional relay oscillator **1212** has the largest product of mass and frequency squared. The couplings between relay oscillators and the coupling between the output oscillator **906** and the first relay oscillator **1216**A, ϕ.sub.r.sub.1, may be turned on at the same time and allowed to evolve thermodynamically according to Langevin dynamics. Once coupling is initiated, each successive relay oscillator takes continuous samples of the previous oscillator it is coupled to. Furthermore, each successive relay oscillator may be a closer approximation of the expectation value of the output oscillator **906**. In this manner, additional relay oscillator **1212** approximates an expectation value of input oscillator **906**. At this point, coupling between the additional relay oscillator **1212** and input oscillator **908** may be turned on and the thermodynamic information may be transferred to input oscillator **908**. The number of relay oscillators and the timing of coupling may be chosen beforehand and optimized for a desired precision or accuracy of the expectation value of the output relay oscillator. For ease of illustration a version that includes bias oscillators is shown. However, it should be understood that in some embodiments bias oscillators may be omitted.

[0155] FIG. **13**A illustrates example couplings between visible neurons of an energy-based model (EBM), according to some embodiments.

[0156] In some embodiments, input neurons and output neurons of an energy-based model, such as visible neurons **1302** and visible neurons **1304**, may be directly linked via connected edges **1306**. As shown in FIG. **13**A, a given visible neuron **1302** of the five shown in the figure is connected, via edges **1306**, to each of the respective three visible neurons **1304**. A person having ordinary skill in the art should understand that FIG. **13**A is meant to represent example embodiments of a graph architecture implemented using a thermodynamic chip that may be applied and that specific numbers of visible neurons **1302** and/or visible neurons **1304** shown in the figure are not meant to be restrictive. Additional configurations combining more/less visible neurons **1302** and/or visible neurons **1304** are also encompassed by the discussion herein. In addition, recall that neurons are logical representations of physical oscillators, such that, when describing neurons in FIGS. **13**A and **13**B, it should be understood that neurons and edges are implemented using oscillators and couplings.

[0157] FIG. **13**B illustrates example couplings between visible neurons and non-visible neurons (e.g., hidden neurons) of an energy-based model (EBM), according to some embodiments.

[0158] In some embodiments, FIG. **13**B may resemble additional example embodiments of an energy-based model architecture implemented using a thermodynamic chip. As shown in the figure, additional non-visible neurons **1308** may be used, which are respectively coupled, via edges **1306**, to both visible neurons **1302** and to visible neurons **1304**. Note that while the non-visible neurons are “not visible” from the perspective of inputs and outputs, the non-visible neurons may each correspond to a given oscillator. In addition, it may be noted that, in some embodiments that make use of non-visible neurons, no direct connections, via edges **1306**, may be implemented between visible neurons **1302** and visible neurons **1304**, but rather connections are routed firstly via non-visible neurons **1308**, as shown in FIG. **13**B. Couplings between visible and non-visible neurons may be additionally referred to herein as “layers” of a given energy-based model architecture that is implemented using a thermodynamic chip, according to some embodiments.

[0159] FIG. **14** is high-level diagram illustrating a process of determining weights and biases to be used in an energy-based model (EBM), wherein the weights and biases are determined using measurement values for synapse oscillators, according to some embodiments.

[0160] As shown in FIG. **14**, in a first evolution, visible neurons of an energy-based model implemented on a thermodynamic chip **1402** may be clamped to input data. For example, multiple mini-batches of input data may be clamped to visible neurons for multiple evolutions used to generate a first set of measurements used to compute a positive phase term. For example, the measurements may be used by classical computing device **1404** to compute the positive phase term.

[0161] Also, in a second (or other subsequent) evolution, the visible neurons may remain unclamped, such that the visible neuron oscillators are free to evolve along with the synapse oscillators during the second (or other subsequent) evolution. Measurements may also be taken and used by the classical computing device **1404** to compute a negative phase term.

[0162] Additionally, the positive and negative phase terms computed based on the first and second sets of measurements (e.g., clamped measurements and un-clamped measurements) may be used to calculate updated weights and biases.

[0163] This process may be repeated, with the determined updated weights and biases used as initial weights and biases for a subsequent iteration. In some embodiments, inferences generated using the updated weights and biases may be compared to training data to determine if the energy-based model has been sufficiently trained. If so, the model may transition into a mode of performing inferences using the learned weights and biases. If not sufficiently trained, the process may continue with additional iterations of determining updated weights and biases.

[0164] FIG. **15** is high-level diagram illustrating a process of determining weights and biases to be used in an energy-based model (EBM), wherein the weights and biases are computed using a classical computing device, according to some embodiments.

[0165] In some embodiments, updated weights and bias values may be computed iteratively by classical computing device **1504** based on inference measurements from thermodynamic chip **1502**. For example, inference values may be compared to training data values, and new weights and biases may be iteratively computed until the inference values closely correspond to the training data. As can be seen in FIG. **15**, in some embodiments the synapse oscillator may be omitted as degrees of freedom of the energy-based model. For example, when a classical computing device is used to iteratively determine the weight and bias values.

[0166] FIG. **16** is high-level diagram illustrating an example neuro-thermodynamic computer comprising a thermodynamic chip (e.g., that implements multiple energy-based models (EBMs) and a relay gadget) included in a dilution refrigerator and coupled to a classical computing device in an environment external to the dilution refrigerator, according to some embodiments.

[0167] In some embodiments, a neuro-thermodynamic computing system **1600** (as shown in FIG. **16**) may be used to implement the various embodiments shown in FIGS. **1-15** and may include one or more thermodynamic chip(s) **1602** placed in a dilution refrigerator **1606**. In some embodiments, classical computing device **1604** may control temperature for dilution refrigerator **1606**, and/or perform other tasks, such as helping to drive a pulse drive to change respective hyperparameters of the given system and/or perform measurements. Also, the classical computing device **1604** may perform other simple computing operations, such as are needed to determine updated weights and biases.

[0168] In some embodiments, classical computing device **1604** may include one or more devices such as a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), and/or other devices that may be configured to interact and/or interface with a thermodynamic chip within the architecture of neuro-thermodynamic computer **1600**. For example, such devices may be used to tune hyperparameters of the given thermodynamic system, etc. as well as perform part of the calculations necessary to determine updated weights and biases. In some embodiments, the classical computing device **1604** may be placed in an environment **1606** outside of the dilution refrigerator **1606**.

[0169] As shown in FIG. **16**, in embodiments where more than one thermodynamic chip is used with a relay gadget, multiple ones of the thermodynamic chips and the relay gadget may be placed in the same dilution refrigerator **1606**.

[0170] FIG. **17** is high-level diagram illustrating an example neuro-thermodynamic computer comprising a thermodynamic chip (e.g., that implements multiple energy-based models (EBMs) and a relay gadget) included in a dilution refrigerator and coupled to a classical computing device that is also included in the dilution refrigerator, according to some embodiments.

[0171] As another alternative, in some embodiments, a classical computing device used in a neuro-thermodynamic computer, such as in neuro-thermodynamic computer **1700**, may be included in a dilution refrigerator with the thermodynamic chip. For example, neuro-thermodynamic computer **1700** includes both thermodynamic chip **1702** and classical computing device **1704** in dilution refrigerator **1706**.

[0172] FIG. **18** is high-level diagram illustrating an example neuro-thermodynamic computer comprising one or more thermodynamic chips (e.g., that implement respective energy-based models (EBMs) and a relay gadget) coupled to a classical computing device in an environment other than a dilution refrigerator, according to some embodiments.

[0173] Also, in some embodiments, a neuro-thermodynamic computer, such as neuro-thermodynamic computer **1800**, may be implemented in an environment other than a dilution refrigerator. For example, neuro-thermodynamic computer **1800** includes thermodynamic chip(s) **1802** and classical computing device **1804**, in environment **1806**. In some embodiments, environment **1806** may be temperature controlled and, the classical computing device (or other device) may control the temperature of environment **1806** in order to achieve a given level of evolution according to Langevin dynamics.

[0174] FIG. **19** is a high-level diagram illustrating oscillators included in a substrate of the thermodynamic chip and mapping of the oscillators to logical neurons of the thermodynamic chip, according to some embodiments.

[0175] In some embodiments, a substrate **1902** may be included in a thermodynamic chip, such as any one of the thermodynamic chips described above. Oscillators **1904** of substrate **1902** may be mapped in a logical representation **1952** to neurons **1954**, as well as weights and biases (shown in FIG. **20**). In some embodiments, oscillators **1904** may include oscillators with potentials ranging from a single well potential to a dual-well potential and may be mapped to visible neurons, weights, and biases.

[0176] In some embodiments, Josephson junctions and/or superconducting quantum interference devices (SQUIDS) may be used to implement and/or excite/control the oscillators **1904**. In some embodiments, the oscillators **1904** may be implemented using superconducting flux elements (e.g., qubits). In some embodiments, the superconducting flux elements may physically be instantiated using a superconducting circuit built out of coupled nodes comprising capacitive, inductive, and Josephson junction elements, connected in series or parallel, such as shown in FIG. **19** for oscillator **1904**. However, in some embodiments, generally speaking various non-linear flux loops may be used to implement the oscillators **1904**, such as those having single-well potential, double-well potential, or various other potentials, such as a potential somewhere between a single-well potential and a double-well potential.

[0177] FIG. **20** is an additional high-level diagram illustrating oscillators included in a substrate of the thermodynamic chip mapped to logical neurons, weights, and biases of a given neuro-thermodynamic computing system, according to some embodiments.

[0178] While weights and biases are not shown in FIG. **19** for ease of illustration, respective ones of the visible neurons **1954** of FIG. **19** may each have an associated bias, and edges connecting the neurons **1954** may have associated weights. Each of the weights and biases may be mapped to oscillators in the thermodynamic chip, as well as the visible (and non-visible) neurons being mapped to oscillators in the thermodynamic chip. For example, FIG. **20** shows a portion of a thermodynamic chip, wherein weights and biases associated with a given neuron **2054** are shown. For example, bias **2056** may be a bias value for visible neuron **2054** and weights **2058** and **2060** may be weights for edges formed between visible neuron **2054** and other visible neurons of the thermodynamic chip. As shown in FIG. **20**, each of the chip elements (visible neuron **2054**, bias **2056**, weight **2058**, and weight **2060**) may be mapped to separate ones of oscillators **2004**. This may allow the visible neurons (and/or hidden neurons), weights, and biases to have independent degrees of freedom within a given thermodynamic chip that can separately evolve.

[0179] In some embodiments, oscillators associated with weights and biases, such as bias **2056** and weights **2058** and **2060**, may be allowed to evolve during a training phase and may be held nearly constant during an inference phase. For example, in some embodiments, larger "masses" may be used for the weights and biases such that the weights and biases evolve more slowly than the visible neurons. This may have the effect of holding the weight values and the bias values nearly constant during an evolution phase used for generating inference values.

[0180] FIG. **21** illustrates an example apparatus for measuring positions of oscillators of a thermodynamic chip using a flux read-out device, according to some embodiments.

[0181] In some embodiments, a resonator with a flux sensitive loop, such as resonator **2104** of flux readout apparatus **2102** may be used to measure flux and therefore position of an oscillator **1504** of thermodynamic chip **122**. Note that flux is the analog of position for the oscillators used in thermodynamic chip **122**. The flux of oscillator **1504** is measured by flux readout device **2102**. For example, if the inductance of oscillator **1504** changes, it will also cause a change in the inductance of resonator **2104**. This in turn causes a change in the frequency at which resonator **2104** resonates. In some embodiments, measurement device **2114** detects such changes in resonator frequency of resonator **2104** by sending a signal wave through the resonator **2104**. The response wave that can be measured at measurement device **2114**, will be altered due to the change in resonator frequency of resonator **2104**, which can be measured and calibrated to measure the flux of oscillator **1504**, and therefore the position of its corresponding neuron or synapse that is coded using that oscillator.

[0182] More specifically, in some embodiments, incoming flux **2106** from resonator **1504** is sensed by the inductor of resonator **2104**, wherein flux tuning loop **2110** is used to tune the flux sensed by resonator **2104**. Flux bias **2108** also biases the flux to flow through resonator **2104** towards transmission line **2112**. In some embodiments, transmission line **2112** may carry the signal outside of a dilution refrigerator, such as dilution refrigerator **1602** shown in FIG. **16**. Also, in some embodiments, transmission line **2112** may carry the signal to a classical computing device located within the dilution refrigerator, such as is shown for dilution refrigerator **1702** in FIG. **17**. Measurement device **2114** may then be used to measure the signal representing the flux and may provide a flux measurement value and/or provide a position measurement value.

[0183] FIG. **22** illustrates an example apparatus for measuring momentums of oscillators of a thermodynamic chip using a charge read-out device, according to some embodiments.

[0184] As mentioned in the discussion of FIG. **21**, flux of an oscillator of the thermodynamic chip corresponds to position. In a similar manner, a charge measurement of an oscillator corresponds to momentum. In some embodiments, a charge or current read out circuit, such as charge or current read out circuit **2202**, may be used to measure charge of a given oscillator of the thermodynamic chip **122**. In such an arrangement, the oscillator **1504** of thermodynamic chip **122** is represented by oscillator **2014**, which is coupled to a SET island **2004** that appears as a small superconducting island from the perspective of the charge or current read out circuit **2202**. For example, the charge or current read out circuit **2202** includes capacitances Ce, Cc, and Cg which are connected in the lower portion of the charge or current read out circuit **2202** as shown in FIG. **22**. The Cg capacitance along with the voltage Vg is used to bias the charge on the SET island. The Ce capacitance along with the voltage V.sub.oscillator, is used to bias the charge of the oscillator **1504**, the Cc capacitance is the capacitance between the SET island **2204** and the oscillator **1504**. The C.sub.set island (e.g. SET island **2204**) is used to measure the charge of the oscillator **1504** with capacitance C.sub.oscillator, since the SET properties (**2208**) are sensitive to the charge on the SET island **2204**, which is coupled to the oscillator charge. The amplifiers (cold and warm) and radio frequency signal source of signal processing **2210** are used to send the measured signal indicating the charge of the oscillator **1504** to a measurement device **2212**, which may be a classical computing device, such as classical computing device **104**.

Illustrative Computer System

[0185] FIG. **23** is a block diagram illustrating an example computer system that may be used in at least some embodiments.

[0186] In some embodiments, the computing system shown in FIG. **23** may be used, at least in part, to implement any of the techniques described above in FIGS. **1**-**22**. Furthermore, computer system **2300** may be configured to interact and/or interface with neuro-thermodynamic computing device **2380**, according to some embodiments.

[0187] In the illustrated embodiment, computer system **2300** includes one or more processors **2310** coupled to a system memory **2320** (which may comprise both non-volatile and volatile memory modules) via an input/output (I/O) interface **2330**. Computer system **2300** further includes a network interface **2340** coupled to I/O interface **2330**. Classical computing functions may be performed on a classical computer system, such as computing computer system **2300**.

[0188] Additionally, computer system **2300** includes computing device **2370** coupled to thermodynamic chip **2380**. In some embodiments, computing device **2370** may be a field programmable gate array (FPGA), application specific integrated circuit (ASIC) or other suitable processing unit. In some embodiments, computing device **2370** may be a similar computing device as described in FIGS. **1**-**22**, such as classical computing devices used to control a thermodynamic chip. In some embodiments, neuro thermodynamic computing device **2380** may be a similar neuro thermodynamic computing device as described in FIGS. **1**-**22**, such as neuro thermodynamic computing devices implemented using thermodynamic chips.

[0189] In various embodiments, computer system **2300** may be a uniprocessor system including one processor **2310**, or a multiprocessor system including several processors **2310** (e.g., two, four, eight, or another suitable number). Processors **2310** may be any suitable processors capable of executing instructions. For example, in various embodiments, processors **2310** may be general-purpose or embedded processors implementing any of a variety of instruction set architectures (ISAs), such as the x86, PowerPC, SPARC, or MIPS ISAs, or any other suitable ISA. In multiprocessor systems, each of processors **2310** may commonly, but not necessarily, implement the same ISA. In some implementations, graphics processing units (GPUs) may be used instead of, or in addition to, conventional processors.

[0190] System memory **2320** may be configured to store instructions and data accessible by processor(s) **2310**. In at least some embodiments, the system memory **2320** may comprise both volatile and non-volatile portions; in other embodiments, only volatile memory may be used. In various embodiments, the volatile portion of system memory **2320** may be implemented using any suitable memory technology, such as static random-access memory (SRAM), synchronous dynamic RAM or any other type of memory. For the non-volatile portion of system memory (which may comprise one or more NVDIMMs, for example), in some embodiments flash-based memory devices, including NAND-flash devices, may be used. In at least some embodiments, the non-volatile portion of the system memory may include a power source, such as a supercapacitor or other power storage device (e.g., a battery). In various embodiments, memristor based resistive random-access memory (ReRAM), three-dimensional NAND technologies, Ferroelectric RAM, magneto resistive RAM (MRAM), or any of various types of phase change memory (PCM) may be used at least for the non-volatile portion of system memory. In the illustrated embodiment, program instructions and data implementing one or more desired functions, such as those methods, techniques, and data described above, are shown stored within system memory **2320** as code **2325** and data **2326**.

[0191] In some embodiments, I/O interface **2330** may be configured to coordinate I/O traffic between processor **2310**, system memory **2320**, computing device **2370**, and any peripheral devices in the computer system, including network interface **2340** or other peripheral interfaces such as various types of persistent and/or volatile storage devices. In some embodiments, I/O interface **2330** may perform any necessary protocol, timing or other data transformations to convert data signals from one component (e.g., system memory **2320**) into a format suitable for use by another component (e.g., processor **2310**). In some embodiments, I/O interface **2330** may include support for devices attached through various types of peripheral buses, such as a variant of the Peripheral Component Interconnect (PCI) bus standard or the Universal Serial Bus (USB) standard, for example. In some embodiments, the function of I/O interface **2330** may be split into two or more separate components, such as a north bridge and a south bridge, for example. Also, in some embodiments some or all of the functionality of I/O interface **2330**, such as an interface to system memory **2320**, may be incorporated directly into processor **2310**.

[0192] Network interface **2340** may be configured to allow data to be exchanged between computing device **2300** and other devices **2360** attached to a network or networks **2350**, such as other computer systems or devices. In various embodiments, network interface **2340** may support communication via any suitable wired or wireless general data networks, such as types of Ethernet network, for example. Additionally, network interface **2340** may support communication via telecommunications/telephony networks such as analog voice networks or digital fiber communications networks, via storage area networks such as Fibre Channel SANs, or via any other suitable type of network and/or protocol.

[0193] In some embodiments, system memory **2320** may represent one embodiment of a computer-accessible medium configured to store at least a subset of program instructions and data used for implementing the methods and apparatus discussed in the context of FIG. **1** through FIG. **22**. However, in other embodiments, program instructions and/or data may be received, sent or stored upon different types of computer-accessible media. Generally speaking, a computer-accessible medium may include non-transitory storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD coupled to computer system **2300** via I/O interface **2330**. A non-transitory computer-accessible storage medium may also include any volatile or non-volatile media such as RAM (e.g., SDRAM, DDR SDRAM, RDRAM, SRAM, etc.), ROM, etc., that may be included in some embodiments of computer system **2300** as system memory **2320** or another type of memory. In some embodiments, a plurality of non-transitory computer-readable storage media may collectively store program instructions that when executed on or across one or more processors implement at least a subset of the methods and techniques described above. A computer-accessible medium may further include transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as a network and/or a wireless link, such as may be implemented via network interface **2340**. Portions or all of multiple computing devices such as that illustrated in FIG. **23** may be used to implement the described functionality in various embodiments; for example, software components running on a variety of different devices and servers may collaborate to provide the functionality. In some embodiments, portions of the described functionality may be implemented using storage devices, network devices, or special-purpose computer systems, in addition to or instead of being implemented using general-purpose computer systems. The term “computer system”, as used herein, refers to at least all these types of devices, and is not limited to these types of devices.

CONCLUSION

[0194] Various embodiments may further include receiving, sending or storing instructions and/or data implemented in accordance with the foregoing description upon a computer-accessible medium. Generally speaking, a computer-accessible medium may include storage media or memory media such as magnetic or optical media, e.g., disk or DVD/CD-ROM, volatile or non-volatile media such as RAM (e.g., SDRAM, DDR, RDRAM, SRAM, etc.), ROM, etc., as well as transmission media or signals such as electrical, electromagnetic, or digital signals, conveyed via a communication medium such as network and/or a wireless link.

[0195] The various methods as illustrated in the figures above and described herein represent exemplary embodiments of methods. The methods may be implemented in software, hardware, or a combination thereof. The order of method may be changed, and various elements may be added, reordered, combined, omitted, modified, etc.

[0196] It will also be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the present invention. The first contact and the second contact are both contacts, but they are not the same contact.

[0197] Various modifications and changes may be made as would be obvious to a person skilled in the art having the benefit of this disclosure. It is intended to embrace all such modifications and changes and, accordingly, the above description is to be regarded in an illustrative rather than a restrictive sense.

Claims

1. A system comprising: one or more controllers configured to initialize one or more thermodynamic chips; and the one or more thermodynamic chips, comprising: a plurality of oscillators, wherein the plurality of oscillators comprises: a set of oscillators configured to generate SoftMax data samples, wherein each SoftMax data sample is a one-hot encoded vector corresponding to a respective energy-based model; one or more input data oscillators which represent input data; a plurality of sets of oscillators, each of which is configured to implement one of the respective energy-based models, which are further configured to process modified input data, wherein a sample of the modified input data has exactly one non-zero vector, and wherein the non-zero vector corresponds to the respective energy-based model which corresponds to the SoftMax data sample; and a set of multiplication oscillators having an engineered potential which causes the set of multiplication oscillators to use the generated SoftMax data samples and the input data as input and generate the modified input data as output.
2. The system of claim 1, wherein: the set of oscillators configured to generate the SoftMax data samples has one or more learnable SoftMax parameters; the one or more learnable SoftMax parameters may be trained to cause the set of oscillators configured to generate the SoftMax data samples to generate one or more inferences using Langevin dynamics; and the one or more inferences are the SoftMax data samples.
3. The system of claim 1, wherein: each of the plurality of sets of oscillators configured to implement respective ones of the energy-based models has one or more learnable energy-based model parameters; and the one or more learnable energy-based model parameters may be trained to cause the plurality of sets of oscillators implementing respective ones of the energy-based models to generate one or more inferences using Langevin dynamics; and the one or more inferences are output data.
4. The system of claim 3, wherein the training is performed: using positive and negative phase terms; or during the initialization by the controller.
5. The system of claim 1, wherein an oscillator of the plurality of oscillators: represents a neuron or a synapse; and wherein an oscillator that represents a synapse: is a weighting value coordination oscillator or a bias value coordination oscillator; and has a weighting or biasing configuration with one or more other oscillators of the plurality of oscillators.
6. The system of claim 1, wherein the set of multiplication oscillators are configured to thermodynamically evolve, based on the engineered potential, to generate the modified input data based on the input data and the SoftMax data samples.
7. The system of claim 1, wherein said initialization comprises clamping the input data to the one or more input data oscillators.
8. One or more thermodynamic chips, comprising: a plurality of oscillators, wherein the plurality of oscillators comprises: a set of oscillators configured to generate SoftMax data samples, wherein each SoftMax data sample is a one-hot encoded vector corresponding to a respective energy-based model; one or more input data oscillators which represent input data; a plurality of sets of oscillators, each of which is configured to implement one of the respective energy-based models, which are further configured to process modified input data, wherein a sample of the modified input data has exactly one non-zero vector, and wherein the non-zero vector corresponds to the respective energy-based model; and a set of multiplication oscillators having an engineered potential, wherein the engineered potential causes the set of multiplication oscillators to use the generated SoftMax data samples and the input data as input and generate the modified input data as output.
9. The one or more thermodynamic chips of claim 8, wherein: the set of oscillators configured to generate the SoftMax data samples has one or more learnable SoftMax parameters; the one or more learnable SoftMax parameters may be trained to cause the set of oscillators configured to generate the SoftMax data samples to generate one or more inferences using Langevin dynamics; and the one or more inferences are the SoftMax data samples.
10. The one or more thermodynamic chips of claim 8, wherein: each of the plurality of sets of oscillators configured to implement respective ones of the energy-based models has one or more learnable energy-based model parameters; and the one or more learnable energy-based model parameters may be trained to cause the plurality of sets of oscillators implementing respective ones of the energy-based models to generate one or more inferences using Langevin dynamics; and the one or more inferences are output data.
11. The one or more thermodynamic chips of claim 10, wherein the training is performed: using positive and negative phase terms; or during the initialization by the controller.
12. The one or more thermodynamic chips of claim 8, wherein an oscillator of the plurality of oscillators: represents a neuron or a synapse; and wherein an oscillator that represents a synapse: is a weighting value coordination oscillator or a bias value coordination oscillator; and has a weighting or biasing configuration with one or more other oscillators of the plurality of oscillators.
13. The one or more thermodynamic chips of claim 8, wherein the set of multiplication oscillators having the engineered potential are configured to thermodynamically evolve, based on the engineered potential, to generate the modified input data based on the input data and the SoftMax data samples.
14. A method, comprising: initializing one or more thermodynamic chips with input data, the one or more thermodynamic chips comprising: a plurality of oscillators, wherein the plurality of oscillators comprises: a set of oscillators configured to generate SoftMax data samples, wherein each SoftMax data sample is a one-hot encoded vector corresponding to a respective energy-based model; one or more input data oscillators, wherein said initialization of the system comprises clamping input data to the one or more input data oscillators; a plurality of sets of oscillators, each of which is configured to implement one of the respective energy-based models, which are further configured to process modified input data, wherein a sample of the modified input data has exactly one non-zero vector, and wherein the non-zero vector corresponds to the respective energy-based model; and a set of multiplication oscillators having an engineered potential, wherein the engineered potential causes the set of multiplication oscillators to use the generated SoftMax data samples and the input data as input and generate the modified input data as output; and receiving output data from the one or more thermodynamic chips.
15. The method of claim 14, wherein the given one of the respective energy-based models generates the output data.
16. The method of claim 14, wherein: the set of oscillators configured to generate the SoftMax data samples has one or more learnable SoftMax parameters; the one or more learnable SoftMax parameters may be trained to cause the set of oscillators configured to generate the SoftMax data samples to generate one or more inferences using Langevin dynamics; and the one or more inferences are the SoftMax data samples.
17. The method of claim 14, wherein: each of the plurality of sets of oscillators implementing respective ones of the energy-based models has one or more learnable energy-based model parameters; and the one or more learnable energy-based model parameters may be trained to cause the plurality of sets of oscillators implementing respective ones of the energy-based models to generate one or more inferences using Langevin dynamics; and the one or more inferences are the output data.
18. The method of claim 17, further comprising: performing the training using positive and negative phase terms; or performing the training during the initialization by the controller.
19. The method of claim 14, wherein an oscillator of the plurality of oscillators: represents a neuron or a synapse; and wherein an oscillator that represents a synapse: is a weighting value coordination oscillator or a bias value coordination oscillator; and has a weighting or biasing configuration with one or more other oscillators of the plurality of oscillators.
20. The method of claim 14, wherein the set of multiplication oscillators having the engineered potential are configured to thermodynamically evolve, based on the engineered potential, to generate the modified input data based on the input data and the SoftMax data samples.